



Bachelor Thesis

Quantization and Spike-Based Explainability in Spiking Neural Networks for Gravitational-Wave Glitch Classification

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Quantization and Spike-Based Explainability in Spiking Neural Networks for Gravitational-Wave Glitch Classification

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Abstract

Quantization makes a neural network cheap enough to deploy, and it is usually validated by checking that accuracy does not fall. On convolutional networks, accuracy has been shown to survive quantization while the explanations of the model’s decisions do not. This thesis asks the spiking analogue: *does a quantized spiking neural network base its decisions on the same evidence as its full-precision counterpart, and at what energy cost?* A spiking network carries a membrane potential that can be quantized independently of the weights, and its explanations are sequences of maps; the working hypothesis is that quantizing the membrane degrades the temporal structure of an explanation before quantizing the weights degrades its spatial structure.

A four-block convolutional spiking network is trained on four classes of transient noise from the LIGO Gravity Spy release, reaching 0.9833 ± 0.0014 validation accuracy over three seeds under hardware-realizable constraints, and its decisions are explained with the gradient-free Spike Activation Map. Before any comparison, the explanation metrics are calibrated: two full-precision models differing only in seed produce maps correlating at 0.153 on the worst class, against a numerical floor of 0.975. Spatial metrics are therefore valid only between a model and its own parent, and every quantized model is fine-tuned from, and compared with, the full-precision model of its own seed.

Weights only, membrane only, and both are quantized at 8, 4 and 2 bits. At 8 and 4 bits accuracy and explanations stay at the level of the parent in every arm. At 2 bits the arms separate: ternary weights collapse the classifier, while a 2-bit membrane keeps accuracy at parity (0.982) and yet changes the explanation, whose highlighted region overlaps the parent’s by 0.44 against a floor of 0.70 — a change the spatial correlation (0.92) would have reported as a preserved explanation. The new maps remain faithful by the deletion metric: they point at evidence the model uses, but not at the evidence the original pointed at. The hypothesis is not supported as an ordering of explanation metrics, but its mechanism is: the 2-bit membrane makes the spike trains of the first layer perfectly periodic, a temporal change the explanation’s temporal metric cannot see. That configuration is also dominated on energy and memory, so there is no trade-off between efficiency and explainability here: a joint 4-bit model keeps both at about one sixtieth of the estimated energy and one eighth of the memory of its parent.

Contents

Abstract	1
1 Introduction	5
1.1 Motivation	5
1.2 Research question	6
1.3 Contributions	6
1.4 Application domain	7
1.5 Structure	7
2 Background and Related Work	9
2.1 Spiking neural networks	9
2.1.1 Training by surrogate gradients	11
2.1.2 Input encoding	11
2.2 Quantization	11
2.2.1 Quantization for spiking networks	12
2.2.2 Energy accounting	12
2.3 Explainability for spiking networks	12
2.3.1 Evaluating an explanation	13
2.4 Gravitational-wave glitch classification	13
2.5 The gap this thesis occupies	13
3 Methodology	15
3.1 Data	15
3.1.1 Acquisition	15
3.1.2 Class selection	15
3.1.3 Feature selection	17
3.1.4 Preprocessing and partition	19
3.2 Model	20
3.2.1 Architecture	20
3.2.2 Readout and loss	20
3.3 Training	20
3.3.1 Hyperparameter optimization	20
3.3.2 Experiment tracking	21
3.3.3 Final training	21
3.4 Selecting the operating configuration	21
3.4.1 Input resolution	21
3.4.2 Hyperparameter search under deployment constraints	22
3.4.3 Baseline manifest	22
3.5 Diagnostics of the network dynamics	22
3.6 The Spike Activation Map	23
3.7 Evaluating and comparing explanations	23

3.7.1	Multi-sample statistics	23
3.7.2	Faithfulness: the deletion metric	24
3.7.3	Divergence metrics	24
3.7.4	Calibration protocol	24
4	The Quantization Study	26
4.1	Design principle: three separate arms	26
4.2	Quantization-aware fine-tuning from the parent	27
4.3	Weight quantizer	27
4.4	Membrane quantizer	28
4.5	Joint arm	29
4.6	Reading the results	29
4.7	Stratified analysis	30
4.8	Energy and memory	30
4.9	Test-set evaluation	31
5	Results	32
5.1	Input resolution	32
5.1.1	Accuracy does not discriminate	32
5.1.2	Network health does discriminate	32
5.1.3	Decision: 112 pixels	33
5.2	Hyperparameters and the full-precision reference	33
5.2.1	β improves the measurement instrument	34
5.2.2	Three-seed reference	34
5.3	Physical validation of the dynamics	34
5.4	What the explanations show	35
5.5	Fixing the explainer	36
5.5.1	Choice of the decay rate	36
5.5.2	Faithfulness	38
5.6	Calibrating the divergence metrics	40
5.6.1	Verification of the instrument	40
5.6.2	Two independent solutions	40
5.6.3	A negligible perturbation	41
5.6.4	What the calibration establishes	42
5.7	Preparing the membrane arm	42
5.7.1	The range of the membrane potential	42
5.7.2	Clipping the potential costs nothing	42
5.8	Energy and memory of the reference	43
5.9	The quantization grid	44
5.9.1	Gates and control	45
5.9.2	Accuracy	45
5.9.3	Explanation divergence	45
5.9.4	The hypothesis	47
5.9.5	Faithfulness	48
5.9.6	Stratification by event strength	49
5.9.7	Energy and memory	49
5.9.8	Still to be measured	50
5.10	Test-set evaluation	50
6	Discussion and Conclusions	51
6.1	The instrument	51
6.2	Quantization and explanation fidelity	52
6.3	Limitations	53
6.4	Future work	54
6.5	Closing remark	55

References	56
A Additional figures	58
B Reproducibility	60

Chapter 1

Introduction

1.1 Motivation

In September 2015 the two Advanced LIGO (Laser Interferometer Gravitational-Wave Observatory) detectors made the first direct detection of a gravitational wave, GW150914, produced by the merger of two black holes. Two years later, LIGO and Virgo jointly observed GW170817, the first signal from a binary neutron star merger. Its electromagnetic counterpart, from a short gamma-ray burst to a kilonova, was then followed by telescopes around the world, marking the beginning of multi-messenger astronomy.

Since then, detections have gone from exceptional events to routine ones, and gravitational-wave astronomy now produces a continuous stream of data that must be analyzed, often in real time. In this setting, the main obstacle is not the faintness of the astrophysical signals but the noise. The detectors are sensitive enough to record a constant stream of short, non-Gaussian transients of instrumental and environmental origin, known as *glitches*. Glitches can mimic genuine signals or overlap with them, which raises the false-alarm rate and reduces the effective sensitivity of searches.

Gravity Spy, the source of the data used in this work, was created to address this problem. By combining citizen-science classification with machine learning, it sorts glitches into morphological families based on their time–frequency spectrograms. Knowing which family a glitch belongs to is the first step towards tracing it back to its physical cause and, where possible, removing that cause at the instrument.

More broadly, machine learning has become indispensable to data-quality work, but it comes at a cost. The demand for computing power is growing faster than the resources available, public alerts must be issued within minutes of a detection, and data volumes will grow further with future observing runs and next-generation detectors. Making models lighter and cheaper to run is therefore a natural goal. Two approaches are particularly relevant: quantization, which compresses an existing network, and spiking neural networks, which are designed from the outset for energy-efficient, event-driven computation.

Running a neural network outside a data center generally means making it smaller, and the standard tool for this is indeed quantization. Quantization represents weights and activations with fewer bits, trading numerical precision for savings in memory and energy. A quantized model is routinely validated by checking that its accuracy has not dropped, and for many applications this is a sufficient test.

It is not sufficient, however, when the model is also expected to explain its decisions. Rabbi et al. [19] evaluated five convolutional architectures at INT8 and INT4, comparing the attribution maps of the quantized models with those of their full-precision counterparts. They found that explanation similarity degrades measurably between INT8 and INT4 even where accuracy differences are negligible. They also found that when quantization does cause a misclassification, the quantized model is generally attending to *different* regions of the input, rather than weighing the same evidence differently.

In other words, accuracy and interpretability do not degrade together.

This finding, however, is limited to conventional artificial neural networks (ANNs). Spiking neural networks (SNNs), the architecture most closely associated with low-power deployment [21], differ from ANNs in ways that make the question both harder and more interesting.

1.2 Research question

The question this thesis addresses is:

Does a quantized spiking neural network under edge-realistic constraints base its decisions on the same evidence as its full-precision counterpart, and at what energy cost?

Three properties of spiking networks have no counterpart in the convolutional setting, and each one bears on the question.

A second quantization target. A spiking neuron carries a membrane potential: a state variable read and written at every time step, and a dominant memory cost at deployment. It is quantized independently of the weights [25, 23], whereas a conventional network offers only weights and activations. The two targets can therefore be separated, which turns a single observation into a causal ablation.

A temporal axis in the explanation. A spiking explanation is a *sequence* of maps, one per time step. Quantization may preserve where a network looks while displacing when it decides, a change to which spatial metrics are blind by construction.

A native explainer. The Spike Activation Map [12] reads spike activity directly. It requires neither gradients, side-stepping the approximation error a surrogate gradient injects into any transplanted Grad-CAM-style method [22], nor ground-truth labels.

The working hypothesis follows from the first two properties:

Quantizing the membrane potential degrades the temporal structure of the explanation before quantizing the weights degrades its spatial structure,

since the membrane potential is the variable that integrates evidence over time, while a weight error is fixed for a whole forward pass. The hypothesis is falsifiable, and testing it requires quantizing the two targets separately: the membrane arm should displace the temporal centre of mass of the explanation at a *higher* bit-width than the one at which the weight arm lowers its spatial correlation. If the order is reversed, the hypothesis is refuted, which is itself a reportable result.

1.3 Contributions

Answering the research question requires an instrument, and the metrics used to compare explanations return numbers between zero and one whatever is fed to them. Before a quantization result can be read, one must know what those metrics report when nothing has changed. The thesis is therefore organized in two parts: the construction and calibration of the instrument, and its application to the quantization grid.

1. **A hardware-realizable full-precision baseline**, with every design choice settled on evidence and recorded in a frozen manifest: the input resolution chosen on network health rather than accuracy, the membrane decay restricted to shift-implementable values inside the hyperparameter search rather than rounded afterwards, and a three-seed reference at 0.9833 ± 0.0014 validation accuracy.

2. **A corrected implementation of the Spike Activation Map**, with its decay parameter fixed on the temporal axis it actually governs, and its faithfulness tested quantitatively against a random baseline rather than judged by visual plausibility.
3. **A calibration of the explanation-divergence metrics** at three levels: numerical noise, a thirtyfold change in the explainer, and initialization variance. Neither of the two closest works [19, 23] establishes any of the three, and the result has the widest scope of the thesis: spatial metrics are valid only between a model and its own parent.
4. **A bit-width characterization of explanation fidelity for a spiking classifier**, with weight and membrane quantization separated and then combined over thirty fine-tuning runs, measured with spatial and temporal divergence metrics read against the calibrated thresholds, tested for faithfulness, reported per class and per seed, and stratified by event strength.
5. **An analytical energy and memory account** of every operating point, so that accuracy, explanation fidelity and energy can be read on a common bit-width axis.

The central result is a configuration that the accuracy criterion cannot tell apart from its neighbours. Quantizing the membrane potential to 2 bits leaves accuracy at parity, 0.982, yet the region its explanations highlight overlaps the full-precision one by 0.44, well below the numerical floor of 0.70; a 4-bit weight model of the same accuracy keeps an overlap of 0.74. The change is real but bounded: the new explanations still point at evidence the model uses, as the deletion metric confirms, just not at the evidence the original model pointed at. The hypothesis stated above is not supported in the form it was written — both arms first move the explanation at the same bit-width, and the 2-bit membrane changes *where* the explanation points more than *when* — but its mechanism is: the same quantization makes the spike trains of the first layer perfectly periodic, a temporal change that the temporal metric of the explanation does not register. The configuration with the changed explanation is also dominated on energy and memory, so in this network there is no trade-off to make: a joint 4-bit model keeps both accuracy and explanation at about 1/60 of the estimated energy and one eighth of the memory.

1.4 Application domain

The classification task is the identification of transient noise — known as *glitches* — in the data of the LIGO gravitational-wave detectors, using the Gravity Spy release [26, 9, 8].

The domain was chosen for three reasons. Explanation fidelity has an *operational* consequence there: glitch classification guides detector characterization, where a commissioner acts physically on the instrument, so a compressed model reaching the same label from different evidence would preserve the label while corrupting the diagnostic cue. The classes have physically interpretable morphology, so a domain expert can judge whether an attention map falls where the glitch actually is, which is not true of a generic image benchmark. And the data is public, large and labelled.

1.5 Structure

Chapter 2 reviews spiking neural networks, their training by surrogate gradients, quantization for spiking architectures, explainability methods applicable to them, and the gravitational-wave classification task. Chapter 3 describes the data pipeline, the model, the selection of the operating configuration, the Spike Activation Map and the protocol used to calibrate the divergence metrics. Chapter 4 describes the quantization study: the quantizers, the grid, the fine-tuning protocol, the divergence metrics and the energy model.

Chapter 5 reports the full-precision reference, the physical validation of the network dynamics, the explainability analyses, the calibration, and the results of the quantization grid. Chapter 6 discusses the findings and their limits, and outlines future work.

Chapter 2

Background and Related Work

2.1 Spiking neural networks

SNNs are often described as the third generation of neural networks. They take their inspiration from biological neurons, which communicate through brief, all-or-nothing electrical impulses called *spikes*. In a conventional ANN, each neuron outputs a continuous value computed once per input. In an SNN, information is carried by sequences of binary events distributed over time: at each of T discrete time steps, a neuron either emits a spike or stays silent. The network therefore processes an input as a short temporal sequence. This holds even when the input is static, as with a spectrogram, which is presented repeatedly or encoded as spikes over the T steps.

This representation is what makes SNNs attractive for low-power computing. Because activations are binary, the multiply–accumulate operations that dominate the cost of an ANN reduce to simple accumulations, performed only when a spike actually arrives. Computation is therefore sparse and event-driven: a silent neuron costs almost nothing. Neuromorphic processors such as Intel’s Loihi are built to exploit this sparsity directly, and the energy advantage of an SNN grows as its firing activity decreases.

The most widely used neuron model in deep SNNs is the leaky integrate-and-fire (LIF) neuron. Each neuron maintains an internal state, its membrane potential U , which integrates incoming input, decays (“leaks”) towards rest over time, and triggers a spike when it crosses a threshold θ .

As shown in Figure 2.1, the LIF model sits between biologically detailed neuron models and the artificial neurons of conventional deep learning.

In the discrete-time formulation used by `snnTorch` [7], the dynamics of a neuron at time step t are

$$U[t] = \beta U[t - 1] + WX[t] - S[t - 1] \theta, \quad (2.1)$$

$$S[t] = \begin{cases} 1 & \text{if } U[t] > \theta, \\ 0 & \text{otherwise,} \end{cases} \quad (2.2)$$

where $WX[t]$ is the weighted input from the previous layer, $S[t]$ is the binary output spike, and the subtraction term resets the potential by θ after each spike. The decay factor $\beta \in [0, 1)$ sets how much of its past state the neuron retains. Values close to 1 give a long memory, so the neuron integrates evidence over many time steps; smaller values make it respond mainly to recent input.

The membrane potential gives the LIF neuron a memory, and this is what separates it from a stateless activation function: the output at time t depends on the entire history of inputs up to t . The same mechanism also creates the main difficulty in training. The spike function in Eq. (2.2) is a step function, whose derivative is zero almost everywhere and undefined at the threshold, so standard backpropagation cannot pass through it. The usual solution is the *surrogate gradient* approach: the step function is kept in the forward pass, but in the backward pass its derivative is replaced with that of a smooth approximation,

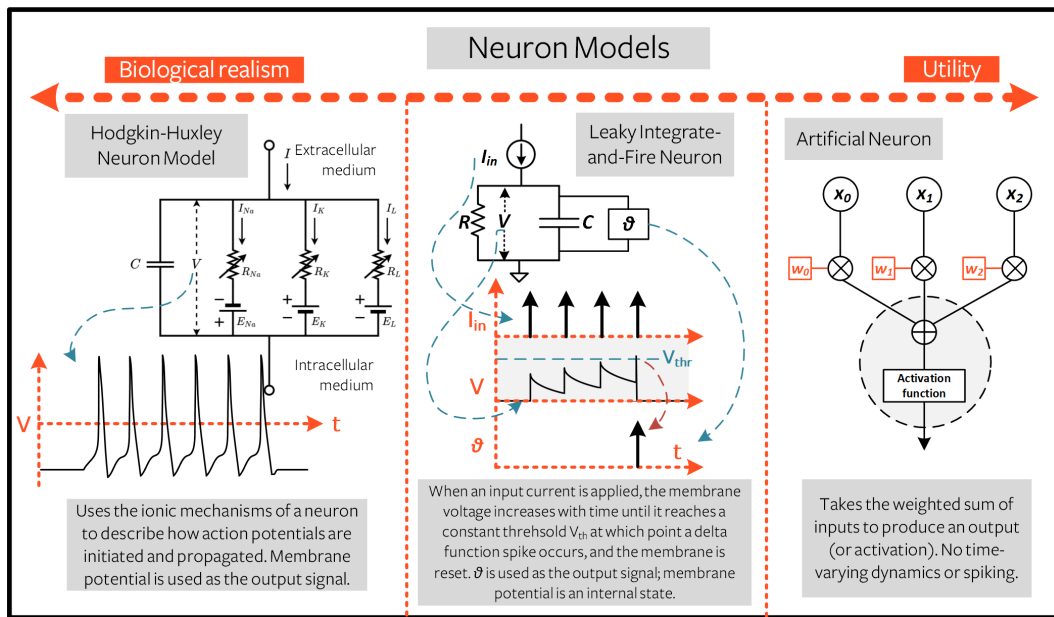


Figure 2.1: Neuron models ordered from biological realism to computational utility. Left: the Hodgkin–Huxley model describes the generation of action potentials through voltage-dependent ionic conductances, at a high computational cost. Centre: the leaky integrate-and-fire (LIF) neuron reduces the membrane to an RC circuit: the capacitor C integrates the input current I_{in} , the resistor R makes the stored charge leak away over time, and a spike is emitted when the membrane potential reaches the threshold (denoted ϑ in the figure and θ in the text), after which the potential is reset. Unlike the Hodgkin–Huxley model, the membrane potential is an internal state and the spike is the output. Right: the artificial neuron of conventional ANNs computes a weighted sum of its inputs followed by an activation function, with no temporal dynamics and no spikes. Reproduced from the snnTorch documentation [6] (MIT License).

such as an arctangent. Combined with backpropagation through time over the T steps, this lets deep SNNs be trained with the same gradient-based tools as conventional networks.

These properties change the quantization problem in three ways. First, SNN activations are already binary, so the precision that quantization removes lies in the weights and in the membrane potential, a state variable with no direct equivalent in an ANN. Second, the threshold makes the network highly sensitive to small numerical changes: a perturbation that moves U slightly above or below θ turns a spike into silence, or the reverse. Because the membrane carries its state forward, such an error does not stay local; it can propagate through later time steps and layers. Third, explanations of an SNN can be temporal as well as spatial. Methods such as Spike Activation Map (SAM) [12] build attribution maps directly from the timing of spikes, without computing gradients, and therefore show not only *where* the network finds evidence in the input but also *when* during the T time steps it does so. Whether quantization preserves this temporal reasoning, and not only accuracy, is the question this thesis sets out to answer.

2.1.1 Training by surrogate gradients

The spike-generating function in Equation 2.1 is a Heaviside step, whose derivative is zero almost everywhere and undefined at threshold. Backpropagation through time is therefore not directly applicable.

The standard remedy is the *surrogate gradient* [16]: the forward pass keeps the exact discontinuous spike, while the backward pass substitutes the derivative of a smooth approximant. The choice of approximant — sigmoid, fast sigmoid, arctangent — affects convergence but not the forward computation, and networks trained this way reach accuracies comparable to conventional architectures on image benchmarks [7].

The substitution has a consequence for explainability that motivates the choice of method in Section 2.3: any gradient-based attribution computed on such a network inherits the approximation error of the surrogate, because the gradients it uses are not the gradients of the function the network actually computes.

2.1.2 Input encoding

Spiking networks require an input that varies over time, while images do not. The two common strategies are *rate coding*, in which pixel intensities become Bernoulli firing probabilities, and *direct* or *constant* encoding, in which the analog frame is presented identically at every time step and the first convolutional layer acts as a learned analog-to-spike converter.

Direct encoding, introduced in this form by DIET-SNN [20], converges in fewer time steps because it does not inject sampling noise into the input. It carries an interpretive consequence that must be stated explicitly: since the stimulus does not vary, **all temporal structure in the network originates from the internal LIF dynamics** — leak, integration and reset — rather than from variation in the input. Every temporal analysis in this thesis is to be read in that light.

2.2 Quantization

Quantization maps a continuous parameter to one of 2^b discrete levels. For a symmetric uniform quantizer over $[-w_{\max}, +w_{\max}]$ the step is

$$\Delta = \frac{2w_{\max}}{2^b - 1}, \quad (2.3)$$

and the maximum rounding error is $\Delta/2$. Equation 2.3 is used in Section 5.6.3 to express a perturbation in units the quantization itself defines. When the quantized quantity is known to be non-negative, an *unsigned* grid over $[0, x_{\max}]$ with step $x_{\max}/(2^b - 1)$ spends

no levels on values that never occur; this is the scheme adopted for the membrane potential in Chapter 4.

The scale can be shared by a whole tensor (*per-tensor*) or computed separately for each output channel (*per-channel*). A per-tensor scale is set by the largest weight in the tensor, so channels whose weights are systematically smaller use only a fraction of the available levels; the per-channel scheme avoids this at the cost of one scale factor per channel [11].

Two families of technique exist. *Post-training quantization* rounds the weights of a trained model, and is what Rabbi et al. [19] apply. *Quantization-aware training* simulates the rounding during training through fake-quantization, and propagates gradients through the non-differentiable rounding operation with a straight-through estimator [4] — the same device, in a different guise, as the surrogate gradient of Section 2.1.1.

2.2.1 Quantization for spiking networks

Spiking architectures add constraints of their own. The membrane decay β multiplies a stored state at every time step, so on integer hardware it is cheapest when realizable as a bit shift: $\beta = 2^{-k}$ is a right shift, and $\beta = 1 - 2^{-k}$ is a shift plus a subtraction. MINT [25] additionally shares a single quantization scale between weights and membrane potentials, which eliminates multipliers from the inference loop entirely.

Smith et al. [23] argue independently that accuracy is an insufficient criterion for quantized spiking networks, and propose an Earth Mover’s Distance between per-neuron firing-rate distributions as a diagnostic of behavioural divergence. They separate weight from membrane quantization as this thesis does, and report that membrane-potential quantization requires at least four bits to preserve both accuracy and dynamics while learned weight quantization remains viable at two. They ground the asymmetry mechanistically: a weight quantization error is fixed for an entire forward pass, whereas a membrane error propagates through the recurrent update at every time step.

2.2.2 Energy accounting

Energy is estimated analytically rather than measured, as is standard in the spiking literature. The count of synaptic operations is multiplied by the per-operation cost at a reference process node, conventionally the 45 nm figures of Horowitz [10]: a 32-bit floating-point multiply-accumulate costs 4.6 pJ against 0.9 pJ for a plain accumulate. That factor of five is the physical reason a spiking network can be cheaper, since a binary activation removes the multiplication, and the accumulation is performed only where a spike occurred.

The same table supplies the integer figures used for quantized operating points: an 8-bit integer multiplication costs 0.2 pJ and an 8-bit addition 0.03 pJ. Narrower integer operations are not tabulated, so any figure below eight bits is an extrapolation and must be declared as such. Memory is accounted separately as the bits needed to store the weights plus the bits needed to store the membrane state of every neuron, which for a spiking network is a cost that a convolutional network of the same shape does not pay.

2.3 Explainability for spiking networks

Attribution methods developed for convolutional networks transfer badly to spiking ones. Gradient-based methods such as Grad-CAM [22] require gradients that, as noted above, exist only as surrogates. Perturbation methods such as LIME are architecture-agnostic but expensive, requiring hundreds of forward passes per explanation.

The Spike Activation Map (SAM) of Kim and Panda [12] was designed natively for spiking networks and requires neither gradients nor labels. It is built on the observation that a neuron firing repeatedly at short intervals is contributing more to the current deci-

sion than one firing sporadically. A *temporal spike contribution score* weights a past spike at t' by its recency,

$$T(t, t') = \exp(-\gamma |t - t'|), \quad (2.4)$$

a *neuronal contribution score* sums those weights over a neuron's preceding spikes, and the map at time t masks that score with the spikes actually emitted at t . The result is one spatial map per time step: an explanation with a temporal axis, which is what makes the question of this thesis answerable at all.

Feature-attribution methods for spiking networks have also been surveyed more broadly [17], but SAM remains the method that reads the spike trains directly.

2.3.1 Evaluating an explanation

An explanation that is stable can still be wrong, so a separate faithfulness test is required. The *deletion metric* of Petsiuk et al. [18] removes pixels in decreasing order of attributed importance and records how quickly the model's confidence falls, comparing against removal in random order. A faithful map degrades the prediction faster than chance.

The metrics used to compare two explanations with each other are borrowed from the saliency-model literature, where they were developed to compare a predicted map against human eye fixations [5]: Pearson correlation, structural similarity [24], and the intersection-over-union of the highest-scoring regions. Bylinskii et al. observe that rankings produced by different metrics frequently disagree, which is the argument for reporting several rather than choosing one.

A separate line of work asks whether attribution methods are testing anything at all. Adebayo et al. [1] randomize model weights progressively and verify that explanations change in response; a map that survives the destruction of the model it purports to explain is not explaining it. The perturbation experiment of Section 5.6.3 is a graded form of that test, run in the opposite direction.

2.4 Gravitational-wave glitch classification

The LIGO interferometers are limited by non-Gaussian transient noise of instrumental and environmental origin. These glitches occur at rates of several per minute, can mimic astrophysical signals, and are grouped into morphological families that correlate with distinct physical causes.

The Gravity Spy project [26, 3] combines machine classification with citizen science to label them. The release used here [8] covers observing runs O1 through O3b and provides, for each Omicron trigger, the trigger metadata, the per-class confidence of a convolutional classifier, the assigned label, and Omega-scan spectrograms.

The classification task is therefore image classification over time–frequency representations. Its interest for this thesis is not that it is difficult — with four well-separated classes it is not — but that the explanations have a consumer. Glitch classification feeds detector characterization, in which the morphology of a glitch family is traced back to an instrumental cause and the instrument is modified. An explanation that points at the wrong part of the spectrogram sends a commissioner to the wrong subsystem.

2.5 The gap this thesis occupies

The two closest works can now be placed precisely. Rabbi et al. [19] measure explanation fidelity under quantization, but for conventional networks, post-training weight-only quantization, and purely spatial explanations. Smith et al. [23] address spiking networks and separate the two quantization targets, but measure divergence in *population firing statistics* aggregated over a dataset, not in explanations.

The unoccupied ground is per-sample, spatially and temporally resolved explanation fidelity for a spiking classifier with the two quantization targets separated. That is the study this thesis performs.

Neither work, however, establishes what its metrics report when nothing has changed. Rabbi et al. present similarity values around 0.85 as evidence of preserved explanations; Smith et al. present EMD values between 0.001 and 0.15 as evidence of behavioural divergence. Chapter 5 shows that for the architecture studied here the corresponding floors are 0.975 for numerical noise and 0.153 for models differing only in initialization, so a value of 0.85 falls between the two and cannot be interpreted without knowing which comparison produced it. Establishing those floors is the contribution with the widest applicability.

Chapter 3

Methodology

This chapter describes the data pipeline, the model, the procedure by which the operating configuration was selected, and the implementation of the Spike Activation Map. Every measurement reported here is on the validation partition.

3.1 Data

3.1.1 Acquisition

The Gravity Spy machine-learning release [8] is distributed as eight CSV files, one per detector (H1, L1) and observing run (O1, O2, O3a, O3b). Each record corresponds to an Omicron trigger with signal-to-noise ratio above 7.5 and peak frequency between 10 Hz and 2048 Hz, and carries the trigger metadata, the per-class confidence of the Gravity Spy classifier, the assigned label, and URLs to Omega-scan spectrograms.

The labels are the classifications produced by that convolutional model, not physical ground truth and not the volunteer-combined label of the separate citizen-science release. Accuracy reported in this thesis is agreement with that reference classifier, which is standard practice on this release and is stated here so the figures are read correctly.

3.1.2 Class selection

Four classes were retained — **Blip**, **Extremely_Loud**, **Scattered_Light**, **Violin_Mode** — on three criteria established before any separability analysis, so that the multivariate checks are a verification rather than the selection rule.

Abundance. These are among the most populated of the 24 classes, with 29 573, 24 002, 99 665 and 2 392 members respectively. The smallest sets the balanced cardinality at 2 392 per class, sufficient to train without augmentation (Figure 3.1).

Morphological diversity. The four span physically distinct families of instrumental noise [26, 9, 3]: short broadband transients of teardrop morphology and ~ 0.04 s duration (Blip); high-amplitude events that saturate the spectrogram (Extremely_Loud); long-duration arches of 2–2.5 s correlated with ground motion, caused in O3 by relative motion between the optical suspension and reaction-mass chains (Scattered_Light); and narrowband lines at ~ 500 Hz and harmonics, corresponding to resonances of the glass fibres suspending the mirrors (Violin_Mode). Spanning the morphological range is what makes the explanation analysis informative, since each class should elicit a recognizably different attention pattern. An example of each is shown in Figure 3.2.

Practical constraint. Four classes keep training and the downstream quantization grid feasible on a single 6 GB consumer GPU.

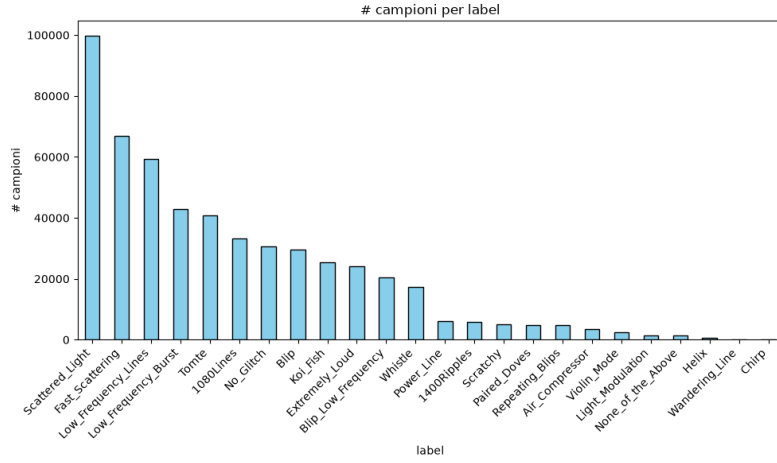


Figure 3.1: Class frequencies in the Gravity Spy machine-learning release, all observing runs and both detectors combined. The distribution is strongly imbalanced; the four retained classes are among the most populated.

SELECTED CLASSES

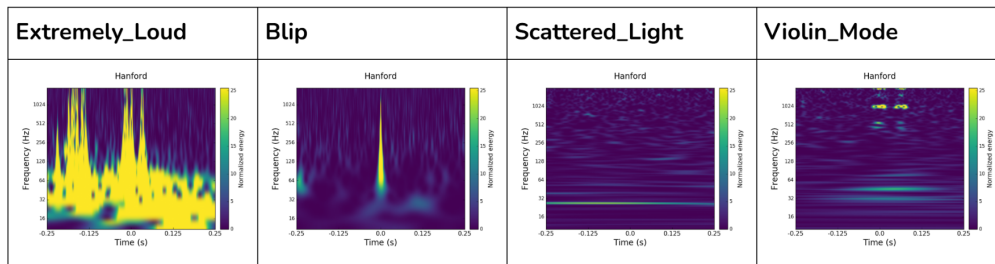


Figure 3.2: One Omega-scan spectrogram per retained class (Hanford detector, 0.5 s window). From left: a saturated broadband event, a short teardrop-shaped transient, a long low-frequency band, and narrow lines near 500 Hz and 1 kHz.

Verification in the tabular domain shows each class isolated by a *different* descriptor, which is the relevant property: Extremely_Loud occupies $\log(\text{SNR} + 1)$ between 6 and 12 while the others lie between 2 and 5; Blip is separated by duration; Scattered_Light sits below roughly 100 Hz in peak frequency with a bimodal bandwidth distribution; and Violin_Mode is the only class extending to 2 kHz (Figure 3.3). A UMAP projection [15] of the standardized descriptors, computed on a stratified subsample of 500 events per class ($n_{\text{neighbors}} = 15$, seed 42), confirms the separation independently (Figure 3.4).

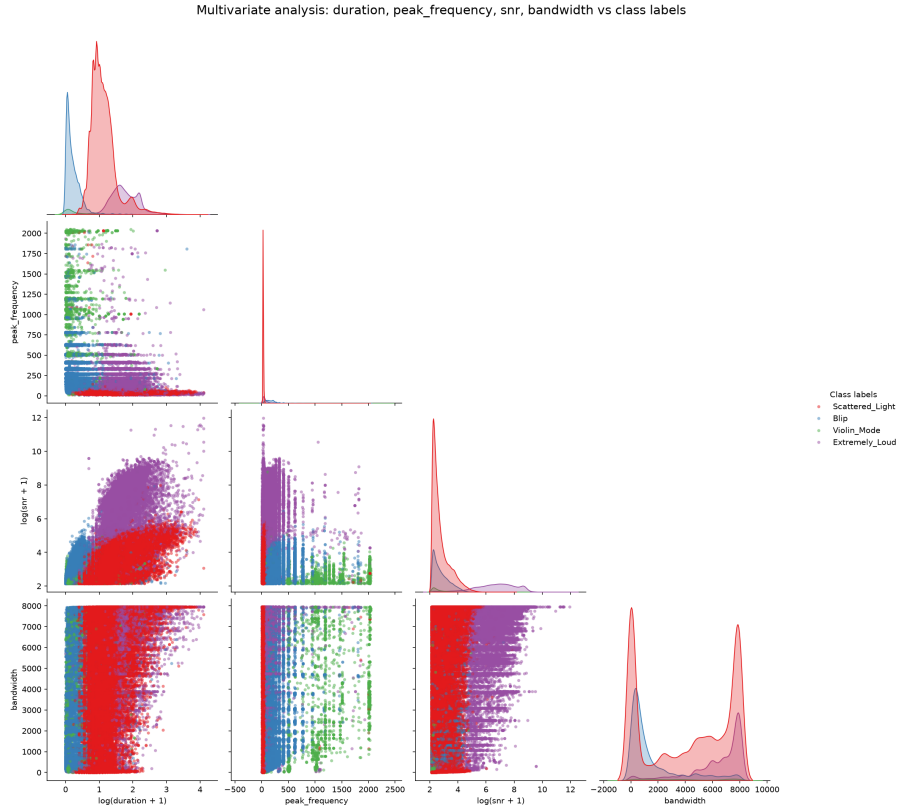


Figure 3.3: Pair plot of the four retained descriptors, coloured by class. Each class is isolated by a different descriptor; Violin_Mode overlaps Blip and Scattered_Light except in peak frequency.

Violin_Mode is the weakest case, and this is stated because it anticipates a result that recurs throughout: in the $\log(\text{SNR} + 1)$ versus $\log(\text{duration} + 1)$ plane it overlaps both Blip and Scattered_Light, being separated only in peak frequency. It is the lowest-recall class at every configuration tested and the most sensitive class in every explainability measurement.

Restricting the problem to four classes keeps it tractable, but means the results are not directly comparable to the full Gravity Spy benchmark.

3.1.3 Feature selection

Of the twelve numerical Omicron descriptors, four were retained on three distinct criteria, read from their distributions and from their correlation matrix (Figure 3.5).

Exact redundancy. The three timestamp variables are mutually correlated at $r = 1.00$, being the same GPS time to within sub-second offsets. Beyond the redundancy, retaining any of them would supply *when* a glitch occurred rather than *what* it is, and the correlation matrix shows this is an exploitable shortcut: event time correlates at -0.45 with the quality factor and -0.22 with peak frequency, reflecting instrumental drift across observing runs.

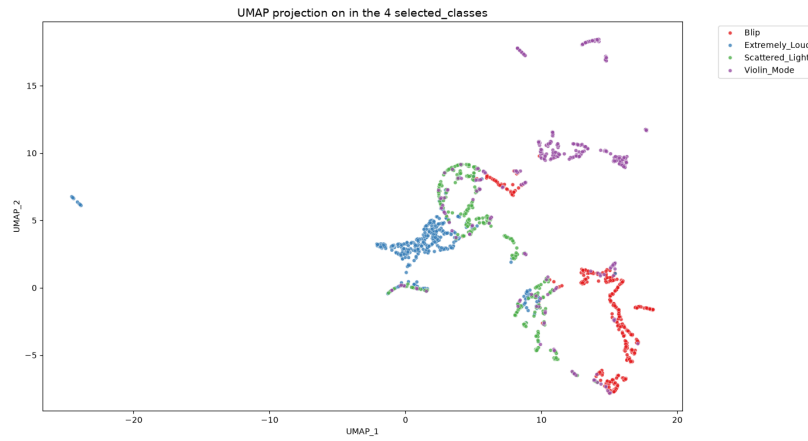


Figure 3.4: Two-dimensional UMAP projection of the standardized descriptors, 500 events per class.

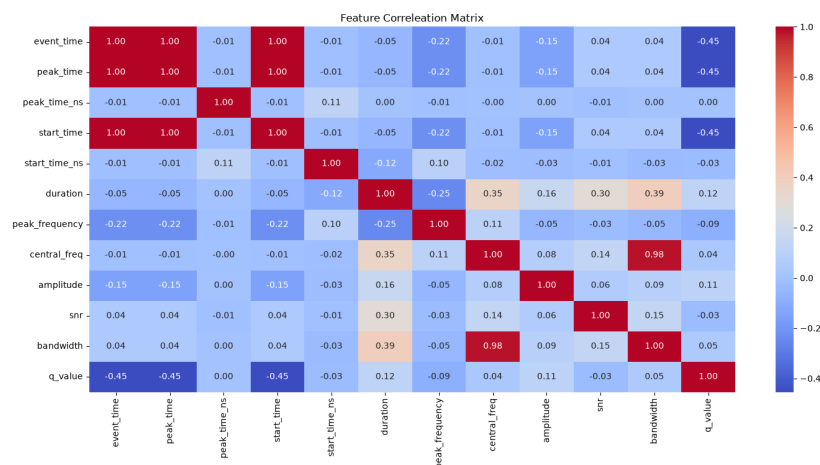


Figure 3.5: Pearson correlation between the twelve numerical Omicron descriptors. The three timestamps are perfectly correlated, and central frequency and bandwidth correlate at 0.98.

Separately, central frequency and bandwidth correlate at 0.98; bandwidth was retained because peak frequency already encodes spectral position, so bandwidth supplies extent rather than duplicating location.

Degenerate distributions. The nanosecond remainders are uniformly distributed and correlate with everything between 0.00 and 0.11, consistent with timing quantization noise. The quality factor contains a sentinel value of order 10^{20} that compresses the real distribution to the axis origin.

Normalized counterpart preferred. Amplitude is a raw strain amplitude, heavily skewed and dependent on the calibration and noise level at the time of the trigger; SNR carries the same information normalized to the local noise floor and is therefore comparable across detectors and epochs.

The four retained descriptors — peak frequency, duration, bandwidth, SNR — form a complete and non-redundant characterization of a transient in the time–frequency plane: spectral position, temporal extent, spectral extent and prominence. Their largest pairwise correlation is 0.39, well below the 0.98 that forced the exclusion of central frequency.

Only one of the four image URL columns was kept. The four correspond to successive time windows of the same detection (0.5 s, 1 s, 2 s, 4 s), and the first was retained. Because they are *different views* rather than duplicates, they are not interchangeable, which is why unavailable images were excluded rather than substituted from another column.

Residual class imbalance is removed by random undersampling to the cardinality of the smallest class, giving an exactly balanced set of 9 568 images.

3.1.4 Preprocessing and partition

The downloaded spectrograms are matplotlib figures comprising a frame with axes, colour bar and annotation. A centre crop of 470×550 pixels isolates the spectrogram panel, which is then resized to the working resolution; the order matters, since cropping the original and resizing afterwards keeps the same portion of the spectrogram framed at every resolution. Channel-wise normalization uses the standard ImageNet statistics, which also fixes the neutral reference value used later by the deletion metric.

The partition is frozen to disk. A dedicated script generates the train/validation/test assignment once, 70 / 15 / 15 under a fixed seed, and writes it as a per-file record of path, class and split; every subsequent run reads that file. A file listed but absent from disk is dropped from its partition with a warning; a file present but unlisted raises, since an unassigned sample would otherwise be used without anyone knowing where it belongs.

Freezing the partition rather than recomputing it gives two properties a seed alone does not. Because the assignment is a function of the file list rather than of the corpus size, removing a file removes exactly that file instead of reshuffling every remaining assignment. And the audit trail is externally verifiable: a reader can confirm that no file appears in two partitions, and the manifest records the SHA-256 digest of the assignment file.

Four downloaded files proved unreadable. Two were recovered by re-downloading; two are permanently unavailable at source (0.02 % of the corpus) and were excluded. **Both belong to the training partition**, so the validation and test sets are complete and identical to those on which the hyperparameters were selected, which is why the search did not need to be repeated.

The final corpus is **9 566 images, partitioned** 6 695/1 435/1 436. Beyond class balance, the three partitions are indistinguishable on the physical properties of the events: median SNR is 14.97/14.09/14.98, with duration, peak frequency and bandwidth similarly matched.

3.2 Model

3.2.1 Architecture

GWGlitchSNN is a four-block convolutional spiking network. Each block is a 5×5 convolution with stride 2 and padding 2 followed by a LIF layer; spatial downsampling is performed by the stride rather than by pooling, which keeps the spike tensors dense in time and avoids inserting a non-spiking operator between layers.

Stage	Operation	Output shape	Notes
Input	direct encoding $\times T$	$[T, B, 3, 112, 112]$	analog frame repeated
Block 1	Conv $3 \rightarrow 16$, k5, s2, p2 + LIF	$[B, 16, 56, 56]$	exposed as <code>spike1</code>
Block 2	Conv $16 \rightarrow 32$, k5, s2, p2 + LIF	$[B, 32, 28, 28]$	exposed as <code>spike2</code>
Block 3	Conv $32 \rightarrow 64$, k5, s2, p2 + LIF	$[B, 64, 14, 14]$	exposed as <code>spike3</code>
Block 4	Conv $64 \rightarrow 128$, k5, s2, p2 + LIF	$[B, 128, 7, 7]$	exposed as <code>spike4</code>
Readout	Flatten (6 272) $\rightarrow$ Linear $\rightarrow$ LIF	$[B, 4]$	one neuron per class

Table 3.1: Layer configuration at the working resolution of 112 px. The flattened width is derived from the input size at construction (25 088 at 224 px, 6 272 at 112 px, 2 048 at 64 px), and the forward pass raises if the input resolution does not match the one the model was built for.

The forward pass returns the stacked output spike train *and* the four hidden spike tensors of shape $[T, B, C, H, W]$, so that every diagnostic and explainability procedure consumes raw hidden spike trains directly from the model interface rather than through hooks.

Parameter count is nearly resolution-independent: 295 332 at 112 px, of which 270 240 are convolutional and unaffected by input size. Only the linear readout shrinks with resolution, a fact with consequences for the design of the quantization grid discussed in Section 4.1.

Batch normalization, pooling and dropout are intentionally absent. Normalization layers interact non-trivially with quantization, and their treatment belongs to a later phase: batch normalization must be reconsidered for spiking networks in general, quantizing its parameters is a problem distinct from quantizing weights and potentials, and two alternative hardware strategies exist — approximating it with shifts, or eliminating it entirely.

3.2.2 Readout and loss

Classification uses a *rate-coded readout*: output spikes are summed over the T time steps and the resulting per-class counts are passed to a cross-entropy loss as logits. Optimization uses Adam [13]. This keeps SAM in the multi-spike regime where its inter-spike-interval formulation remains meaningful.

3.3 Training

3.3.1 Hyperparameter optimization

Hyperparameters are searched with Optuna [2], maximizing validation accuracy. Each trial reports validation accuracy after every epoch and is pruned by a Hyperband pruner [14] (minimum resource 1 epoch, maximum 15, reduction factor 3), which allocates epochs to promising configurations and truncates the others early. A handler catches out-of-memory errors, releases the model and optimizer and marks the trial as pruned, so that configurations with a long time window can fail on the 6 GB budget of the available

GPU without interrupting the study. Studies are persisted to SQLite, one study name per search and resolution.

Two studies were run. The first (v4), at 224 px, searched the learning rate in $[10^{-4}, 3 \times 10^{-3}]$ (log-uniform), β in $(0, 1)$ and $T \in \{5, \dots, 15\}$, and selected a learning rate of 1.036×10^{-4} , $\beta = 0.4148$ and $T = 7$; this configuration was used for the resolution study. The second (v5), at the working resolution, imposes the deployment constraints described in Section 3.4.2 and produced the configuration used throughout the rest of the thesis.

3.3.2 Experiment tracking

Every trial is logged to Weights & Biases. Beyond validation loss and accuracy, the logged metrics include the mean firing rate of each of the four hidden layers. Firing rate is the proxy for energy consumption in spiking hardware [21], so tracking it per layer during the search shows whether high-accuracy configurations achieve their accuracy through dense and expensive activity or through sparse spiking, and whether any layer is silent or saturated before any explainability diagnostic is run.

A separately logged quantity, recorded under the name `avg_firing_rate`, is the mean over the four *output* neurons only. A value near 0.25 indicates that roughly one output neuron fires consistently, which is a signature of confident classification rather than a measure of network activity. It is *not* the global firing rate of the network and is never used for energy accounting, for which only the per-hidden-layer rates are valid.

3.3.3 Final training

The training script reads the frozen configuration and trains from scratch, parametrized on input resolution and seed, so that the same code serves the resolution study, the final training and the multi-seed reference. The seed is set explicitly before the model is constructed. The checkpoint is written whenever validation accuracy improves, and training stops after five consecutive epochs without improvement in validation loss, with a ceiling of 50 epochs. Each run writes the checkpoint together with a record of the full per-epoch history, the per-layer firing rates, the per-class validation recall, the seed and the resolution.

All experiments run on a single NVIDIA RTX 4050 Laptop GPU (6 GB). One epoch over the balanced training partition takes approximately three minutes at 224 px and 45 seconds at 112 px.

3.4 Selecting the operating configuration

Three choices define the operating point: the input resolution, the three hyperparameters, and the decay constant of the explainer. Each is settled on evidence and recorded in a single frozen manifest. Taking any of them on test-set performance would consume the split reserved for the final estimate, so all three are decided on validation.

3.4.1 Input resolution

Nine training runs were performed: three resolutions (224, 112, 64 px) at three seeds each, on the frozen partition, with hyperparameters held fixed and only the input size varying. Since those hyperparameters had been found at 224 px, the protocol systematically handicaps the lower resolutions, and any deficit they show is an upper bound.

The procedure and its outcome are reported in Section 5.1.

3.4.2 Hyperparameter search under deployment constraints

The search uses Optuna [2] with a Hyperband pruner [14], maximizing validation accuracy over three hyperparameters chosen because they govern three distinct axes: learning rate (optimization stability), β (membrane leak and temporal memory), and T (latency, computation and memory cost).

Two deployment constraints are imposed inside the search space rather than by rounding the result. β is restricted to $\{0.25, 0.5, 0.75, 0.875\}$, values whose decay is realizable as a bit shift, and the set spans both families ($\beta = 2^{-k}$ and $\beta = 1 - 2^{-k}$) deliberately. T is capped at 8 for the same reason. A rounded value is one that was never validated; constraining the search guarantees that whatever it returns is deployable by construction, and Section 5.2 shows that in this case the two differ.

The lower bound of the learning rate was moved from 10^{-4} to 10^{-5} , because the v4 optimum (1.036×10^{-4}) sat on the boundary of the earlier range and a boundary optimum is not an optimum.

3.4.3 Baseline manifest

The outcome of these choices is recorded in a single file, `baseline_manifest.json`: the input resolution, the three hyperparameters, the originating study, the decay rate and reference layer of the explainer, the list of seeds (1234, 2345, 3456), the checkpoint paths, the size of each partition, and the first sixteen hexadecimal digits of the SHA-256 digest of the partition file (754b2515b6509075). Every downstream script reads the model and its configuration from the manifest rather than from hard-coded paths, so that no analysis can silently run on a different model or a different partition.

3.5 Diagnostics of the network dynamics

Two diagnostics characterize whether a trained network is a healthy spiking system or a formally correct one whose neurons are dead or saturated. Both are reused unchanged on every quantized model.

Dead and saturated neurons. For each hidden layer, the mean firing rate of every individual neuron is accumulated across the validation partition, weighted by batch size. A rate below 0.01 identifies a *dead* neuron, above 0.80 a *saturated* one. Dead neurons are wasted capacity and, being permanently silent, cannot appear in any explanation; saturated neurons carry almost no information, a nearly-always-firing binary unit having minimal entropy, while consuming the maximum possible energy.

Regularity of firing. The coefficient of variation of the inter-spike intervals,

$$CV = \frac{\sigma_{ISI}}{\mu_{ISI}}, \quad (3.1)$$

distinguishes regular, clock-like firing ($CV \rightarrow 0$) from irregular, Poisson-like firing ($CV \approx 1$). It is relevant here because the Spike Activation Map is built on inter-spike intervals. A random sample of 200 neuron positions per layer is fixed once, at the first batch, and the intervals of exactly those positions are accumulated over every validation image; the CV is computed only where at least five intervals were observed. Since a neuron can produce at most $T - 1 = 7$ intervals within a single image, pooling over the whole partition is what makes the estimate stable.

3.6 The Spike Activation Map

SAM is implemented in two stages, following the original formulation [12]. The temporal spike contribution score weights a past spike at t' by its recency with respect to the current step t :

$$T(t, t') = \exp(-\gamma |t - t'|). \quad (3.2)$$

The neuronal contribution score of a neuron at time t sums that weight over the neuron's preceding spikes, so a unit that has recently fired repeatedly — one with short inter-spike intervals — receives a high score:

$$N_{ij,t}^k = \sum_{t' \in P_{ij}^k, t' < t} T(t, t'). \quad (3.3)$$

The heat map at time t masks that score with the spikes actually emitted at t and sums over the channel axis, yielding one spatial map per time step:

$$M_{ij,t} = \sum_k N_{ij,t}^k S_{ij,t}^k. \quad (3.4)$$

The summation range is exclusive of t . This is stated explicitly because an earlier implementation in this work summed over $t' \leq t$, and the error is instructive. The score measures the contribution of a neuron's *preceding* trajectory; including $t' = t$ adds a term of weight $\exp(0) = 1$ for every neuron firing at t , so the map produced is the correct SAM *plus the raw binary spike map*. The pedestal is independent of both γ and the spike history, and therefore inflates every correlation between maps computed at different γ . A consequence of the correct definition is that SAM at $t = 0$ is identically zero, no past history existing at the first step.

The implementation uses the recursion

$$N_{t+1} = e^{-\gamma} (N_t + S_t), \quad N_0 = 0, \quad (3.5)$$

algebraically identical to Equation 3.3 but $O(T)$ rather than $O(T^2)$. A reference implementation computing the explicit sum is retained and the two are asserted equal at run time.

Visualizations overlay every time step on the denormalized input under a *shared colour scale*, so that a change in intensity across time is genuine and not an artefact of per-panel rescaling.

The reference configuration for every analysis below is **layer 2 with $\gamma = 0.5$** , both recorded in the manifest. The choice of γ is justified in Section 5.5.1.

3.7 Evaluating and comparing explanations

3.7.1 Multi-sample statistics

A single heat map is a glimpse, not a description of what the network attends to for a class. For each class, SAM is computed on 25 validation images, aggregated over time (mean over T) and stacked, and three maps are reported: the *mean*, the region attended on average; the *standard deviation*, where the explanation varies most across instances; and their ratio, the *coefficient of variation*, which indicates whether a region is attended consistently or only in some examples. Mean and standard deviation share a colour scale across classes so that they are numerically comparable; the coefficient of variation has an extremely long tail where the mean is near zero, and its colour scale is truncated at the 95th percentile.

The same statistics are repeated for all four hidden layers, on 20 images per class. Since SAM sums over the channel axis and the four layers have 16, 32, 64 and 128 channels, raw magnitudes are not comparable across depth, and both maps are normalized by the channel count of their layer before being plotted on a shared scale.

3.7.2 Faithfulness: the deletion metric

Stability is not correctness, so faithfulness is tested separately with the deletion metric of Petsiuk et al. [18]:

1. the time-aggregated SAM map of an image is upsampled bilinearly from the layer resolution to the input resolution, so that importance is expressed over input pixels;
2. the class predicted on the unmodified image is recorded and tracked throughout, since SAM is class-agnostic and the metric measures faithfulness to the model's decision;
3. pixels are removed in decreasing order of SAM score, in 21 steps from 0 % to 100 %, and the softmax confidence of the tracked class is recorded at each step;
4. the identical procedure is repeated with pixels removed in random order, as a control;
5. the curves are averaged over 15 images per class and compared through their area under the curve (AUC), computed by the trapezoidal rule.

A faithful map degrades the confidence faster than random removal, so $\Delta = \text{AUC}_{\text{SAM}} - \text{AUC}_{\text{random}} < 0$ is the expected outcome. Deleted pixels are set to zero in normalized space, which corresponds to the dataset mean rather than to black; the ordering is computed once per image and sliced progressively, so that the removed sets are nested for both curves.

3.7.3 Divergence metrics

Two explanations of the same image, produced by two models, are compared with three metrics, each answering a different question.

Spatial Pearson correlation (PCC). The linear correlation between the two time-aggregated maps, flattened. It is invariant to an affine rescaling of either map, which is the property that makes it usable across models whose overall spiking activity differs.

Top-k intersection over union (IoU). The overlap between the sets of the $k = 20\%$ highest-scoring positions of the two maps. Two independent random rankings overlap at a chance level of $k/(2 - k) = 0.111$.

Temporal centre of mass (CoM). The mean time step at which the energy of the SAM map is concentrated,

$$\text{CoM} = \frac{\sum_t t E_t}{\sum_t E_t}, \quad E_t = \sum_{i,j} M_{ij,t}, \quad (3.6)$$

reported as the absolute difference between the two models. Being an aggregate over the whole map, it is the metric that carries the temporal axis.

In addition, the *per-timestep correlation* between the maps at each t localizes in time where agreement breaks down.

3.7.4 Calibration protocol

The divergence metrics are calibrated before they are used, at three levels, all on fifteen validation images per class at the reference layer and γ , on the three checkpoints of the full-precision reference.

1. **Verification.** A checkpoint compared with itself must return perfect agreement; two calls to the sampling routine must return bit-identical images; and two forward passes of the same model must return identical maps.
2. **Independent solutions.** The three full-precision checkpoints are compared pairwise, which measures how similar two independently trained solutions are.
3. **Numerical floor.** Gaussian noise is added to the weight tensors of one checkpoint, at magnitudes expressed as fractions of its measured 8-bit quantization step, and the perturbed model is compared with the unperturbed one. Biases and the LIF buffers are excluded, since perturbing β would mix membrane dynamics with weight noise.

The γ sensitivity analysis supplies a further reference point: the divergence produced by changing the explainer itself thirtyfold.

Chapter 4

The Quantization Study

This chapter describes how the quantized models are built and how they are compared with their full-precision parents. Every design choice below is either inherited from the calibration of Section 5.6 or settled by a measurement reported in Section 5.7; none is taken on the test partition.

4.1 Design principle: three separate arms

The critical design choice is to **quantize the two targets separately before quantizing them together**. Quantizing weights and membrane potential at once would produce a single observation from which the contribution of each could not be recovered; separating them turns the grid into a causal ablation and is what makes the hypothesis of Chapter 1 testable.

Arm	Weights	Membrane potential	Purpose
FP32 reference	32-bit float	32-bit float	baseline for every comparison
FP32 control	32-bit float	32-bit float	same fine-tuning, quantizer inert
Weight-only (W)	8 / 4 / 2 bit	32-bit float	isolates the synaptic (spatial) effect
Membrane-only (U)	32-bit float	8 / 4 / 2 bit	isolates the state (temporal) effect
Joint (W+U)	8 / 4 / 2 bit	8 / 4 / 2 bit	deployable configuration

Table 4.1: The quantization grid: nine quantized configurations and one control, each at the three seeds of the reference (30 fine-tuning runs).

The *control* arm runs exactly the same fine-tuning procedure with no quantizer attached. It answers the question of whether the pipeline itself — reloading, fine-tuning, checkpoint selection, re-evaluation — changes the model or its explanations when nothing is quantized.

Two quantities are deliberately left out of the grid. The membrane decay $\beta = 0.75$ is already shift-implementable, having been constrained inside the hyperparameter search (Section 3.4.2), and is not re-quantized, which would be a third, unplanned intervention. The biases are kept in full precision: there are 244 of them, under 0.1 % of the parameters, and on integer hardware they are added once to a wide accumulator.

The two arms have *opposite* sensitivities to the resolution choice, which independently justifies treating them as separate ablations. Because 270 240 of the 295 332 parameters are convolutional and therefore resolution-independent, the weight arm achieves nearly the same absolute memory reduction at any resolution, whereas the membrane state falls twelvefold between 224 and 64 pixels.

4.2 Quantization-aware fine-tuning from the parent

Quantization-aware training is implemented by fake quantization: in the forward pass each quantized tensor is rounded to its grid and clamped to the grid’s bounds, and in the backward pass the rounding is replaced by the identity, the straight-through estimator [4]. The weight quantizer is attached as a parametrization of each weight tensor, so that the optimizer updates the underlying full-precision weights while the forward pass only ever sees their quantized values; the membrane quantizer is attached as a forward hook on each hidden LIF layer.

Every quantized model is fine-tuned from the full-precision checkpoint of the same seed.

This is the constraint that determines the rest of the design. The numerical floor of the calibration ($PCC = 0.975$) holds for comparisons between a model and its own parent, where no channel permutation exists; a quantized model trained from a random initialization would be an independent solution, for which the relevant level is 0.153 and almost no quantization effect would be distinguishable from initialization variance.

Schedule. The fine-tuning is short and at reduced learning rate, adapting the parent to its grid rather than finding a new solution: Adam at one tenth of the reference learning rate (1.047×10^{-5}), at most 20 epochs, early stopping after 5 epochs without improvement of the validation loss. Before the first epoch the model is evaluated with the quantizer attached and no training, which gives the *post-training quantization* (PTQ) accuracy of the same configuration at no extra cost; that un-fine-tuned model is saved as the first candidate checkpoint, and it is replaced only by an epoch that strictly improves the validation accuracy. The selected checkpoint is the one on which every subsequent measurement — accuracy, explanations, firing statistics, energy — is taken. The 30 runs took 13 hours on the GPU of Section 3.3.

Gates before the grid. Two checks were run before any fine-tuning. The first verifies the quantizer itself: attaching and removing it restores the original weights exactly, and folding the quantized values into a plain checkpoint and reloading it leaves no drift (0 at 4 bits). The second is the PTQ gate: the joint 8-bit model, with no fine-tuning, must match its parent to within the spread of the full-precision reference, four samples out of 1435. At 8 bits the weight step is of the order of the perturbation that left the spatial correlation at 0.975 in the calibration, so an appreciable loss of accuracy there would indicate a defect of the implementation rather than a physical effect. Both gates passed (Section 5.9.1).

Execution order. All 8-bit configurations were run first, then 4-bit, then 2-bit, so that a defect would surface on the configurations where nothing is expected to happen. The runner skips any configuration whose result is already on disk.

4.3 Weight quantizer

Weights are quantized on a *symmetric signed* grid with $q_{\max} = 2^{b-1} - 1$ levels on each side of zero, so that zero is exactly representable: 255 levels at 8 bits, 15 at 4 bits and 3 at 2 bits (the values $-\Delta, 0, +\Delta$, a ternary network). The quantizer is applied to every convolutional and linear weight tensor, including the readout.

The scale is computed per output channel. The choice was made on the measured weight distributions of the reference checkpoint (Table 4.2). In the deeper convolutions the largest weight of the weakest channel is less than half the largest weight of the whole tensor, so a per-tensor scale would leave that channel using fewer than half of the levels: about one effective bit lost, which at 2 bits is the difference between a ternary channel and a channel quantized entirely to zero.

Layer	$\max w $, tensor	$\max w $, weakest channel	Ratio	Bits lost per-tensor
conv1	0.144	0.115	1.25	0.33
conv2	0.126	0.060	2.11	1.08
conv3	0.113	0.054	2.11	1.08
conv4	0.099	0.044	2.26	1.18
linear	0.214	0.146	1.46	0.55

Table 4.2: Weight ranges of the reference checkpoint (seed 1234). “Bits lost” is $\log_2$ of the ratio: the resolution that the weakest channel would forgo under a single per-tensor scale.

The global 8-bit step used in the calibration (0.00168, from the largest absolute weight of the network, 0.214) is therefore an upper bound on the per-channel steps actually applied, whose mean is between 0.0005 (conv4) and 0.0013 (readout) at 8 bits.

4.4 Membrane quantizer

The membrane potential of the four hidden LIF layers is quantized at every time step. The quantizer acts on the potential returned by each layer after the spike decision, so the quantized value is the *state* carried into the next step: what is stored, and therefore what costs memory, is what is quantized. The output layer, four neurons, is left in full precision; its state is negligible in memory and quantizing it would act directly on the logits.

A grid for the membrane cannot be designed without knowing the range of the quantity, and unlike a weight, the potential has no range until the network is run. Its distribution was therefore measured over the whole validation partition (Section 5.7), and the clipping range was chosen by testing candidate ranges on the full-precision network before any quantization.

The membrane grid is unsigned over $[0, 2]$, twice the firing threshold $U_{\text{thr}} = 1$, with 2^b levels starting exactly at zero. The step is $2/(2^b - 1)$:

Bits	Levels	Step	Step / threshold
8	256	0.0078	0.8 %
4	16	0.1333	13.3 %
2	4	0.6667	66.7 %

Table 4.3: The membrane grid at each bit-width. At 2 bits the representable potentials are 0, 2/3, 4/3 and 2: a single level lies strictly between rest and threshold.

Zero is the value to which every neuron is initialized and around which it rests, so it must be exactly representable; a grid whose levels missed zero would bias every resting neuron by a fraction of a step at every time step. The unsigned grid guarantees it by construction, its first level being zero.

Table 4.3 contains a quantitative prediction. At 8 bits the rounding error is below one percent of the threshold; at 4 bits it is an eighth of it; at 2 bits the error is of the same order as the threshold itself, and a neuron’s state can encode only whether it is at rest, part-way to threshold, or above it. This is the mechanism behind the observation of Smith et al. [23] that membrane quantization needs at least four bits, and the grid tests it directly.

The unsigned scheme is a consequence of the clipping study rather than an assumption: the network turns out to make no use of negative potentials, so a signed grid would spend half of its levels on values that carry no information, the equivalent of losing one bit at every bit-width.

4.5 Joint arm

The joint arm applies both quantizers at once, at the same bit-width, with the per-channel weight scales of Section 4.3 and the fixed membrane grid of Section 4.4. **The two scales are independent.** MINT [25] shares a single scale between weights and membrane so that the inference loop needs no rescaling multiplier; that choice is an optimization of a specific integer datapath, and imposing it here would couple the two targets that the grid exists to separate. With independent scales the joint arm is exactly the composition of the two single-target arms, so any interaction between them shows up as a departure from their combined effect rather than being built into the quantizer.

4.6 Reading the results

For every quantized model, SAM maps are recomputed at the reference layer and γ on twenty validation images per class, and compared with those of **the full-precision parent of the same seed**, using the metrics of Section 3.7.3: spatial PCC, top-20% IoU, the temporal centre of mass (absolute and signed shift), and the per-timestep correlation curve.

Three further measurements characterize behaviour rather than explanations:

- **cross-faithfulness:** the deletion metric applied to the quantized model with its *own* map and with the map of its *parent*, on ten validation images per class and eleven deletion fractions. The second is the operational question, since in practice one is tempted to explain a deployed model with the maps of the original. The deletion baseline is recomputed for the parent of each seed, and a class is interpretable for a seed only if its baseline gap between SAM and random deletion is at least 0.05;
- the **diagnostics** of Section 3.5, collected for every model in a single pass over the validation partition: per-layer firing rate, dead and saturated fractions, CV-ISI, and per-sample predictions;
- the **Earth Mover's Distance** between the per-neuron firing-rate distributions of quantized model and parent, for direct comparison with Smith et al. [23].

In the single validation pass the CV-ISI is computed for every neuron rather than for a sample of 200, and the saturation threshold is 0.90 rather than 0.80; its values are therefore compared only across configurations, never with those of Section 5.3.

Five rules, all established in Chapter 5 before the grid was run, govern how those numbers are read.

1. **Below the numerical floor is noise.** A divergence smaller than that produced by weight noise of one 8-bit step (PCC 0.975, IoU 0.704, CoM difference 0.029 steps, worst class) is not a result.
2. **IoU is an early indicator, PCC a graded measure.** IoU has a dynamic range of $2.6\times$ its floor and saturates early; PCC has $34\times$.
3. **Everything is reported per class and per seed.** The sensitivity ordering Violin_Mode > Scattered_Light > Blip > Extremely_Loud recurs under every intervention tested, so a pooled figure would average a class that responds strongly with one that does not respond at all.
4. **A flat result on Extremely_Loud is uninformative.** That class did not register any perturbation tested; absence of effect and inability to detect one cannot be told apart.
5. **Cross-faithfulness is interpretable only where the parent has a baseline signal.** On the reference seed that means Blip and Extremely_Loud; since the baseline is recomputed per seed, the set of interpretable classes is itself reported.

A sixth rule protects all of them. Every quantized model is compared with *its own* parent, never with the parent of another seed. Because the three parents differ from each other far more than any quantized model differs from its own (Section 5.6), a comparison against the wrong seed would produce a large, spurious divergence. The guard against that error is a property every correct comparison must show: the 8-bit weight arm, the smallest intervention in the grid, must keep a spatial correlation above 0.98 with its parent in every seed.

The test of the hypothesis. The hypothesis predicts an *ordering*: the membrane arm should displace the temporal centre of mass beyond its floor at a higher bit-width than the one at which the weight arm lowers the spatial correlation beyond its floor. The reverse order refutes it, and so does a simultaneous change.

4.7 Stratified analysis

An aggregate divergence per configuration admits only a yes-or-no answer. Joining the per-sample outputs to the physical descriptors allows the result to be reported as a relation instead.

The hypothesis is that divergence increases as event strength decreases, with a mechanism specific to spiking networks: quantizing the membrane potential flips a neuron’s spike decision only where the potential lies within the quantization error of the threshold, and weak events produce potentials that sit close to threshold rather than overshooting it. Because a flipped spike is a discrete event that propagates both to the next layer and forward in time through the decay and reset, and because SAM is constructed from spike trains, the flip changes the object the explanation is made of rather than merely perturbing it.

The prediction discriminates between arms: membrane quantization applies a fixed absolute error and should show a steep dependence on event strength, whereas weight quantization perturbs the input current approximately in proportion to the signal and should be flatter. It is tested on the membrane arm at 4 and 2 bits, the configurations in which the arm has an effect and accuracy is intact.

Two preconditions were verified. SNR does translate into image intensity, the Gravity Spy colour scale being fixed rather than auto-scaled per image. And SNR is very nearly a proxy for class — the top global SNR quartile is 92.8 % *Extremely_Loud* — so stratification uses **terciles computed within each class**, fifteen images per class and tercile, or any trend measured would be a class effect wearing a strength label. The summary statistic is the IoU gradient, high-SNR tercile minus low-SNR tercile; a positive gradient means that weak events lose their explanation more.

4.8 Energy and memory

Synaptic operations are counted per layer as

$$\text{SOP}_\ell = f_{\ell-1} \cdot T \cdot \text{MAC}_\ell, \quad (4.1)$$

where $f_{\ell-1}$ is the mean firing rate of the preceding hidden layer and MAC_ℓ the multiply-accumulate count of the equivalent dense layer, and multiplied by the per-operation costs of [10]. The rates are the per-hidden-layer figures measured on the *selected checkpoint* of each configuration, in the same validation pass that produces its diagnostics, so that energy, accuracy and explanations all refer to the same model; never the output-layer quantity logged as `avg_firing_rate`.

One layer is asymmetric. Under direct encoding the first convolution receives analog pixel values rather than spikes, so its operations are multiply-accumulates. Because the

input is identical at every time step, the current it produces is constant in time and is computed once rather than T times. This is a standard implementation of constant encoding, but stating it matters: recomputing that layer at every step would make the spiking network more expensive than its convolutional equivalent (Section 5.8).

Per-operation costs. Full-precision configurations, the membrane arm included, use 4.6 pJ per MAC and 0.9 pJ per accumulate; configurations with quantized weights use the 8-bit integer figures, 0.2 and 0.03 pJ. Narrower integer operations are not tabulated, so at 4 and 2 bits the costs are extrapolated: an accumulate scales linearly with the operand width and a multiplication quadratically, giving 0.015 and 0.05 pJ at 4 bits and 0.0075 and 0.0125 pJ at 2 bits. These two rows are estimates and are marked as such wherever they appear. The first convolution is costed at the weight bit-width, which assumes that the input image is quantized to the same width; Section 5.9.7 shows how much the conclusion depends on that assumption.

Memory is the number of bits needed to store the 295 088 weights plus those needed to store the membrane state of the 94 080 hidden neurons at the working resolution.

4.9 Test-set evaluation

The test partition is opened once, after every design decision has been taken, on the full-precision reference and on the quantized configurations selected for the final comparison. Accuracy is reported with a 95 % Wilson interval and per class, and is computed by two routes — an aggregate over batches and a per-sample classification — asserted equal.

Chapter 5

Results

5.1 Input resolution

5.1.1 Accuracy does not discriminate

Input size	Val. accuracy	Std. dev.	Observed range	Epochs
224 px	0.9854	0.0007	0.9847–0.9861	22.7
112 px	0.9800	0.0035	0.9777–0.9840	17.3
64 px	0.9814	0.0011	0.9805–0.9826	24.7

Table 5.1: Validation accuracy by input resolution, mean and standard deviation over three seeds.

The ordering in Table 5.1 is not monotonic — 64 px outperforms 112 px — and a genuine resolution effect would produce monotonic degradation. The arithmetic agrees: 0.9854 corresponds to 21.0 errors out of 1 435 validation samples and 0.9814 to 26.6, a gap of 5.6 samples, against a 95 % Wilson interval on a single measurement of ± 0.7 percentage points. The four classes survive an elevenfold reduction in pixel count, consistent with their being distinguished by coarse morphology rather than fine texture.

Per-class recall is directional but small. Violin_Mode degrades most from 224 px to 64 px, by 1.00 percentage points, which is the direction predicted from morphology since narrow spectral lines are the structure most vulnerable to downsampling — but the magnitude is 3.6 samples out of 365, inside the confidence interval of a single measurement. Extremely_Loud degrades more at 112 px than at 64 px, a second non-monotonicity. **No class collapses at reduced resolution.**

5.1.2 Network health does discriminate

Since accuracy is uninformative, the resolution was decided on the internal state of the network. For each hidden layer, the mean firing rate of every individual neuron is accumulated across the validation partition. A rate below 0.01 identifies a *dead* neuron, above 0.80 a *saturated* one. Both are pathological in different ways: dead neurons represent wasted capacity and, being permanently silent, cannot appear in any explanation; saturated neurons carry almost no information, a nearly-always-firing binary unit having minimal entropy, while consuming the maximum possible energy.

Layer 4 is the only layer that responds to resolution, and it responds on two measures at once: the dead fraction falls monotonically from 36.3 % to 19.5 %, while the median per-neuron firing rate rises from 0.0399 to 0.1499. These were the two competing explanations for the higher mean activity observed at lower resolution — more neurons participating,

Input size	Layer 1	Layer 2	Layer 3	Layer 4
224 px	12.1 ± 3.3	16.6 ± 4.9	12.6 ± 2.3	36.3 ± 4.0
112 px	17.8 ± 3.4	11.8 ± 0.8	13.0 ± 1.0	23.6 ± 2.9
64 px	16.4 ± 4.8	11.2 ± 2.3	11.7 ± 2.3	19.5 ± 1.5

Table 5.2: Percentage of dead neurons by layer and resolution, mean $\pm$ standard deviation over three seeds, on validation.

or the same neurons firing harder — and both occur. Saturation does not increase to compensate: neurons above 0.80 remain below 2.5 % in layer 4 at every resolution.

The interpretation is spatial redundancy, not a cured pathology. Layer 4 contains 25 088 neurons at 224 px and 6 272 at 112 px, so in absolute terms the number of *live* neurons falls from roughly 15 970 to 4 789. Reducing the resolution does not increase usable capacity; it reduces capacity while preferentially discarding units that were contributing nothing. At 224 px the deepest layer views a 14×14 map derived from a largely empty spectrogram, and a third of its units never encounter signal.

5.1.3 Decision: 112 pixels

Most of the benefit comes from the first halving: 224 to 112 px recovers 12.7 percentage points of dead fraction, the further move to 64 px only 4.1 more. Membrane state at 8 bits falls from 368 KiB to 92 KiB, bringing the full inference state within the SRAM budget of the target class of devices.

The decisive argument against 64 px is the spatial resolution of the explanations themselves. SAM maps are computed on layer activations, whose side lengths are 56, 28, 14 and 7 at 112 px but 32, 16, 8 and 4 at 64 px. A 4×4 map contains sixteen values, on which centre-of-mass, top-K intersection-over-union and structural similarity are all meaningless. Since the contribution of this thesis is the fidelity of explanations rather than of predictions, a configuration in which the measured object is degenerate in one layer cannot be accepted whatever it saves in compute.

5.2 Hyperparameters and the full-precision reference

The constrained search selected $\beta = 0.75$, $T = 8$, **learning rate** 1.047×10^{-4} at a best validation accuracy of 0.9847 within the 15-epoch budget.

The constraint on β is justified empirically, not only by argument. An unconstrained search over $\beta \in (0, 1)$ selects 0.4148, whose nearest admissible neighbours are 0.25 and 0.5. The constrained search selects 0.75. Rounding the unconstrained optimum after the fact — the alternative to constraining the space — would therefore have produced a configuration that is both inferior and never validated.

All four values were sampled; the search allocated 40 % of trials to 0.75 and 10 % to 0.875, which is itself the surrogate model’s judgement after early exploration. Among trials surviving at least three epochs, 0.75 leads on both the best (0.9847) and the median (0.9763) validation accuracy, while 0.875 trails on both and was pruned in all three of its trials.

$T = 8$ is an optimum at the imposed ceiling and is reported as such: the search was forbidden from exploring $T > 8$. The mitigating evidence is that an unconstrained search with range up to 15 selected $T = 7$, so the preference for long integration windows is mild.

5.2.1 β improves the measurement instrument

Holding resolution and partition fixed and changing only β from 0.4148 to 0.75, the layer-4 dead fraction is *unchanged* at 23.6 %. β does not affect the dead fraction; resolution does. This strengthens the reading of Section 5.1: the deep-layer problem is spatial redundancy rather than a dynamical pathology curable by a better decay constant.

What β does change is saturation, by an order of magnitude and most where it matters. Neurons above 0.80 fall from 9.3/3.5/2.7/1.8 % across the four layers to 4.7/0.22/0.06/0.14 %. The layer-2 figure is the consequential one, because a saturated neuron accumulates the maximum possible contribution score on every image and therefore contributes a constant, non-discriminative floor to every SAM map. **Selecting β on validation accuracy alone happened to reduce that contamination by a factor of sixteen at the reference layer.**

5.2.2 Three-seed reference

Quantity	Mean	Std
Validation accuracy	0.9833	0.0014
Epochs to early stop	21.0	1.7
Recall, Blip	0.9915	0.0028
Recall, Extremely_Loud	0.9855	0.0000
Recall, Scattered_Light	0.9839	0.0027
Recall, Violin_Mode	0.9726	0.0099

Table 5.3: Full-precision reference at 112 px, $\beta = 0.75$, $T = 8$, over three seeds on validation.

The reference is statistically indistinguishable from the 224 px configuration (0.9854): three samples out of 1435 separate them, against a four-sample spread across seeds. **Reducing the pixel count by a factor of four, and constraining β and T to hardware-realizable values, costs no measurable accuracy.**

Accuracy is reproducible; internal dynamics are not. At fixed accuracy the mean firing rate of layer 1 varies by a factor of 2.6 across seeds, and the coefficient of variation falls monotonically with depth: 46.9 % at layer 1, 28.6 % at layer 2, 15.3 % at layer 3, 9.5 % at layer 4. The regime is established within the first epoch and persists, indicating that initialization selects the firing regime and training does not subsequently equalize it. A measurement taken during the resolution study puts the effect in the units of the closest work: the Wasserstein-1 distance between the per-layer firing-rate profiles of two full-precision models differing only in seed is 0.039, inside the range of Earth Mover’s Distances that Smith et al. [23] report *between full-precision and quantized models*. This is the observation that motivates Section 5.6.

5.3 Physical validation of the dynamics

At the adopted configuration the confusion matrix diagonal reads 0.99/0.99/0.98/0.97, consistent with Table 5.3; the only non-negligible errors are on Violin_Mode, the class that in the tabular domain partially overlaps both Blip and Scattered_Light (Figure 5.1).

The dead fraction is 16.1/12.6/11.3/22.8 % across the four layers, and **no neuron in any layer exceeds the saturation threshold**, a consequence of $\beta = 0.75$ (Figure 5.2). These figures reproduce those of Section 5.1 exactly for the same seed, computed through an independent code path, which is a free verification of both.

One feature deserves note: the layer-1 distribution is **bimodal**, with a secondary population near a firing rate of 0.65–0.75 that has no counterpart in the deeper layers. A

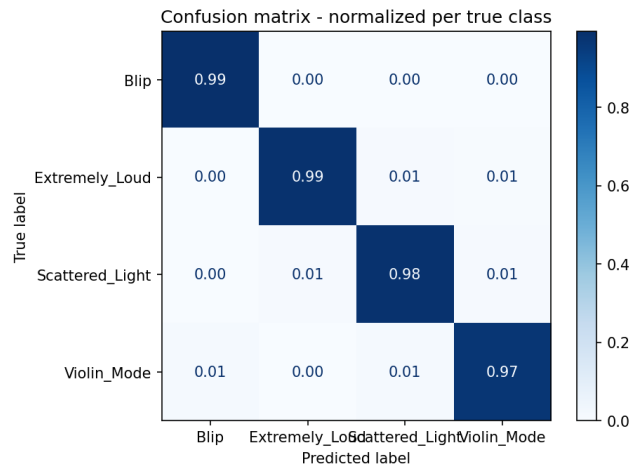


Figure 5.1: Confusion matrix of the reference checkpoint (seed 1234) on validation, normalized per true class.

high-activity subpopulation therefore exists in the first layer only, which may be related to its being the layer with the largest coefficient of variation across seeds.

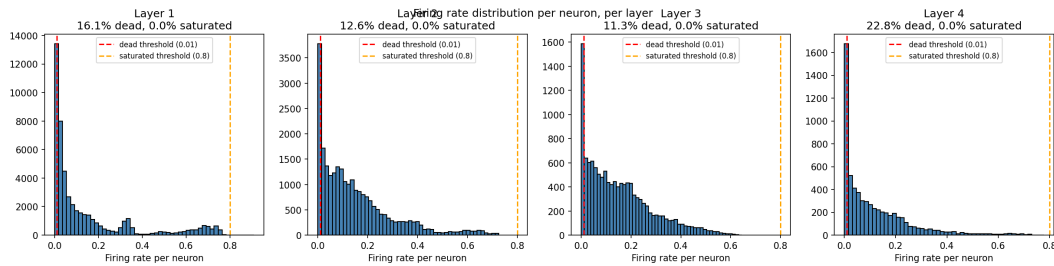


Figure 5.2: Distribution of per-neuron firing rates in the four hidden layers, on validation, with the dead (0.01) and saturation (0.80) thresholds.

The coefficient of variation of inter-spike intervals concentrates between 0.35 and 0.45 in all four layers. Firing is regular rather than Poisson-like, which is the expected outcome under constant direct encoding: since the input does not vary over time, any irregularity observed is generated entirely by the membrane dynamics and by interaction between layers. This is evidence about internal dynamics, not a claim of biological realism. With $T = 8$ a neuron can produce at most seven intervals, so the five-interval threshold used for the estimate is close to the ceiling, which is why the accumulation runs over the whole partition rather than a single image. Between 184 and 196 of the 200 sampled neurons per layer met the threshold (Figure 5.3).

5.4 What the explanations show

Figure 5.4 shows the SAM sequence of a single Blip at the reference layer. The map is zero at $t = 0$ by definition and builds up over the first steps as spike histories accumulate; from the third step onward the attended region is the vertical stroke of the glitch.

The class statistics (Figure 5.5) show four recognizably different attention patterns, each matching the morphology of its class. For Blip the mean map concentrates on the narrow vertical stroke and so does the variability, which is the stroke shifting slightly between instances. For Extremely_Loud the attended region is large and saturated, covering

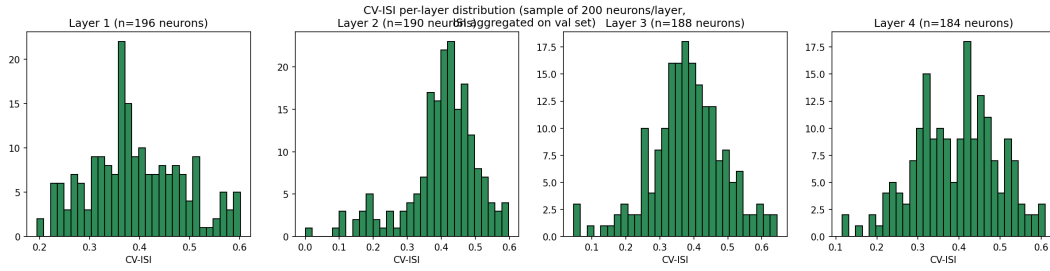


Figure 5.3: CV of inter-spike intervals for 200 sampled neurons per layer, accumulated over the validation partition.

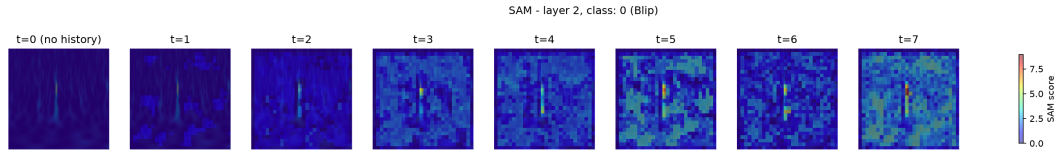


Figure 5.4: SAM at layer 2 for one Blip, one panel per time step, on a shared colour scale.

the lower and central part of the spectrogram. For *Scattered_Light* the attention lies on a horizontal band at low frequency. *Violin_Mode* produces the weakest and most diffuse mean map, with variability concentrated in a few localized spots along the line structure; it is also the class with the lowest recall, and the one whose explanations prove most sensitive in every test below.

Across depth (Figure 5.6) the progression is the expected one, from fine localization in the early layers to coarse localization in the deep ones, with maps of side 56, 28, 14 and 7. At layer 4 a 7×7 map leaves only 49 positions, too few for a top-20% overlap or a structural comparison to be meaningful, while layer 2 still resolves the morphology of every class.

TO DO. Confirm (or replace) this as the reason for choosing layer 2 as the reference layer; the progress report records the choice but not its justification.

5.5 Fixing the explainer

5.5.1 Choice of the decay rate

SAM has one free parameter, and an explanation that changed qualitatively with an arbitrary choice of γ would not be trustworthy.

γ is the decay of a temporal kernel, so it must be evaluated on the temporal axis. Comparing maps averaged over time removes precisely the axis on which γ acts, and a high spatial correlation is then close to guaranteed regardless of γ . Three quantities are therefore measured: the spatial Pearson correlation between time-aggregated maps, retained for continuity with the literature; the *temporal centre of mass* of the SAM energy; and the *per-timestep correlation* against a reference γ .

Before a null result could be trusted, the instrument was checked analytically: on a fully firing spike train with $T = 8$ the centre of mass moves from 4.86 at $\gamma = 0.1$ to 4.02 at $\gamma = 3.0$, so the metric does respond. A second check comes free from the definition: at $t = 1$ the score is $e^{-\gamma} S_0$, so changing γ multiplies the map by a constant, and Pearson correlation is affine-invariant. The measured per-timestep correlation at $t = 1$ is exactly 1.000 for every γ , as it must be.

Two conclusions follow, and neither is the one the naive analysis would reach.

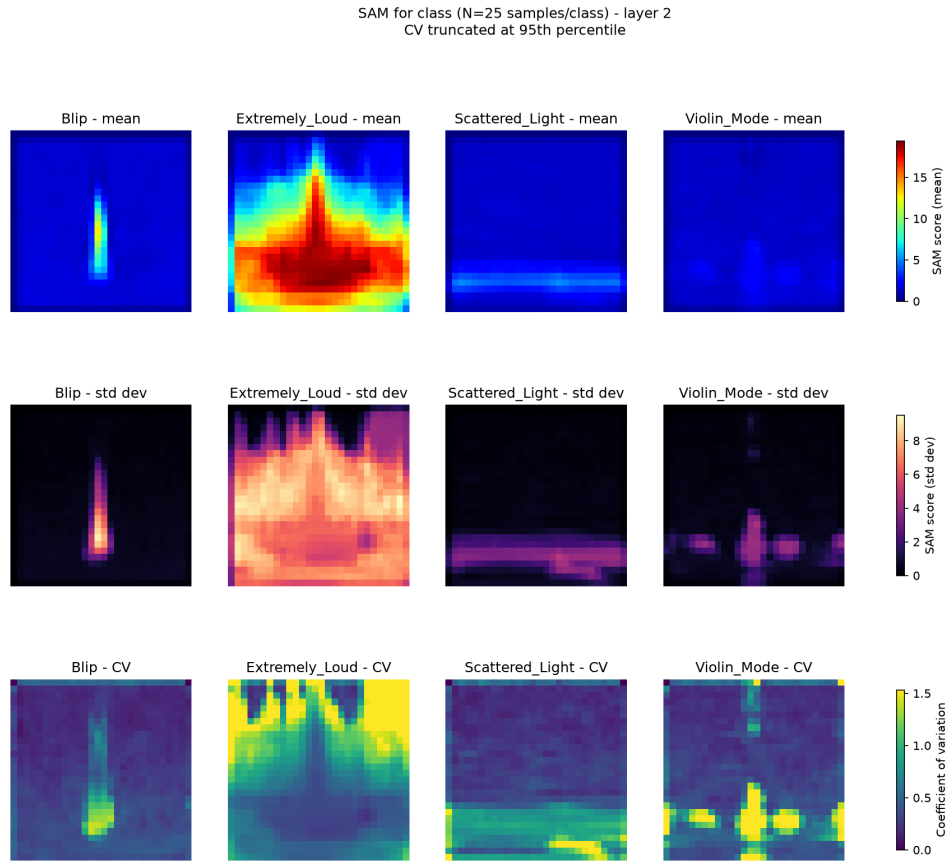


Figure 5.5: SAM statistics per class at layer 2, 25 validation images per class: mean (top), standard deviation (middle) and coefficient of variation (bottom, truncated at the 95th percentile). Mean and standard deviation share a colour scale across classes.

Class	Spatial PCC, γ 0.1 vs 3.0	CoM at 0.1	at 3.0	Shift
Extremely_Loud	0.9993	4.973	4.144	0.829
Blip	0.9890	5.192	4.339	0.853
Scattered_Light	0.9758	5.271	4.352	0.919
Violin_Mode	0.9598	5.260	4.327	0.933

Table 5.4: γ sensitivity at layer 2, mean over fifteen validation images per class.

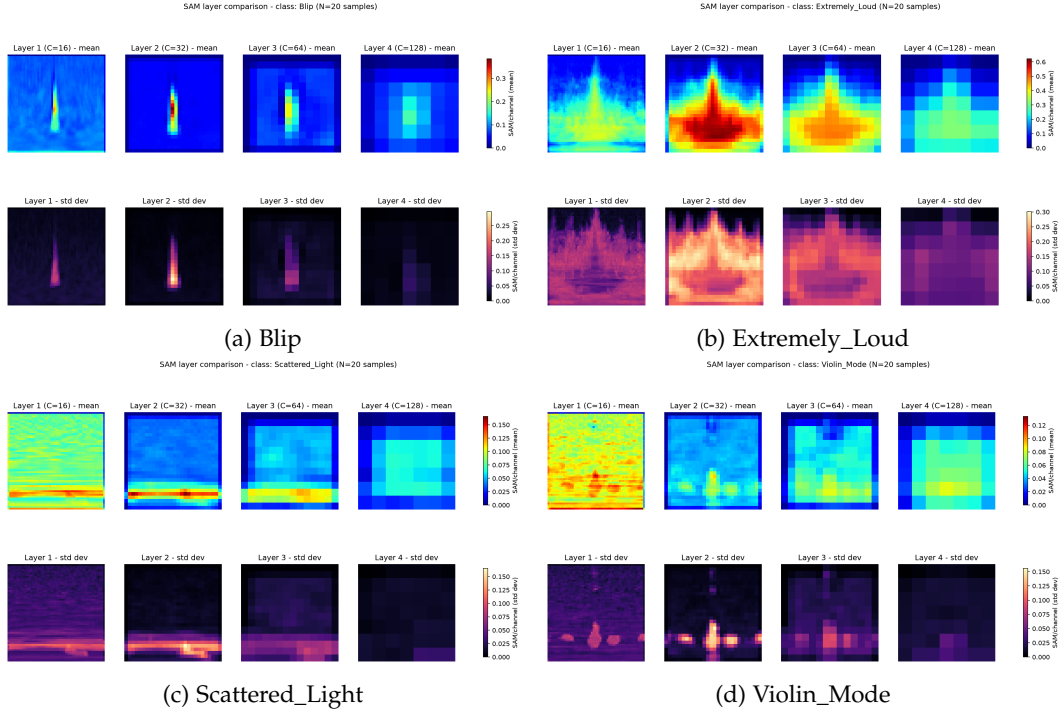


Figure 5.6: SAM mean and standard deviation across the four hidden layers, 20 images per class, normalized by the channel count of each layer.

The temporal shift is largely kernel arithmetic rather than explanation structure. The measured shift of 0.83–0.93 steps matches what a *fully firing* spike train produces (0.84), whereas the layer-2 firing rate is approximately 0.19 and simulated random trains at that density shift by only 0.26 steps. SAM is therefore dominated by a minority of near-continuously firing neurons, for which the centre of mass follows the exponential kernel. The centre of mass against γ is consequently *not* a useful criterion for selecting γ — though it remains valid as a divergence metric, where models are compared at fixed γ and the trivial component cancels, and it now carries a reference scale.

Sensitivity is strongly class-dependent. In terms of $1 - \text{PCC}$, Violin_Mode is 57 times more sensitive than Extremely_Loud. The ordering Violin_Mode > Scattered_Light > Blip > Extremely_Loud recurs identically under the two unrelated interventions of Section 5.6.

Within $\gamma \in [0.3, 0.9]$ all per-timestep correlations remain above 0.98 even in the worst class; only $\gamma = 3.0$ diverges appreciably. $\gamma = 0.5$ is **adopted**, the value of the original formulation [12], now supported by evidence on the axis γ actually governs (Figure 5.7).

5.5.2 Faithfulness

Robustness to γ is necessary but not sufficient: a stable explanation can be stably wrong. The deletion metric [18] removes pixels in decreasing order of SAM score and records the confidence assigned to the class predicted on the unmodified image, comparing against removal in random order.

Three implementation details govern interpretation. Deleted pixels are set to zero *in normalized space*, corresponding to the dataset mean rather than to black, a neutral substrate that avoids injecting an artificial edge. The same map is used for both orderings. And the ordering is computed once per image and sliced progressively, so the masked sets are nested: what is removed at 50 % stays removed at 55 %. Regenerating a random permutation at each fraction would break nesting for the baseline while leaving it intact for SAM, making the two curves incomparable.

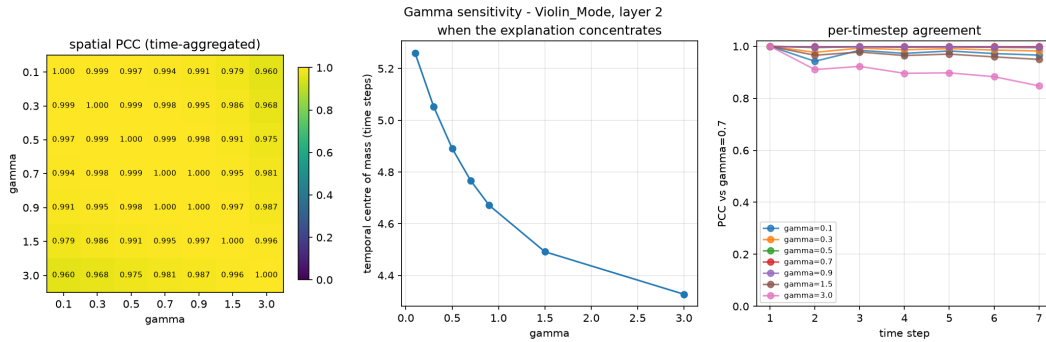


Figure 5.7: γ sensitivity for Violin_Mode, the most sensitive class, at layer 2. The equivalent figures for the other three classes are in Appendix A.

Class	AUC-SAM	AUC-random	Δ	Relative	Reading
Blip	0.3119	0.4795	-0.1676	-35.0 %	faithful
Extremely_Loud	0.6121	0.8892	-0.2771	-31.2 %	faithful
Scattered_Light	0.3835	0.3650	+0.0185	+5.1 %	not separable
Violin_Mode	0.4528	0.4540	-0.0012	-0.3 %	indistinguishable

Table 5.5: Deletion metric by class, 15 validation images each. A faithful explanation gives AUC-SAM below AUC-random, so a negative Δ is expected.

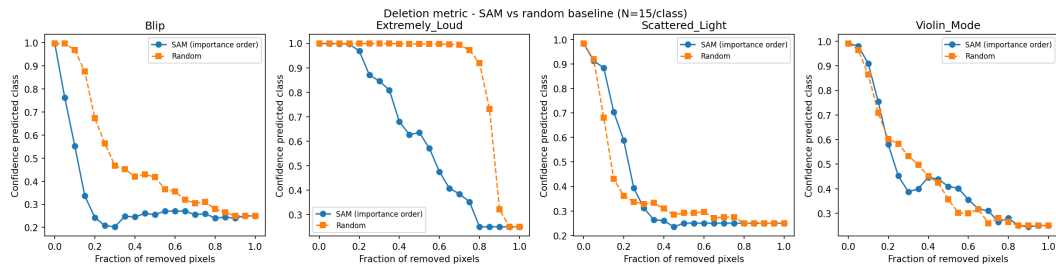


Figure 5.8: Deletion curves, SAM order against random order, 15 validation images per class.

The metric separates SAM from the baseline on two classes and not on the other two, and the split is systematic. Blip and Extremely_Loud show reductions of 35 and 31 % in area. Violin_Mode is within 0.3 % of the baseline. Scattered_Light is nominally inverted, but at +5.1 % the magnitude is one sixth of the two positive effects, so it is better described as an absence of separation with a slight drift in the wrong direction than as a genuine reversal.

The pattern follows the spatial geometry of the classes rather than any property of SAM. Blip is a compact vertical line and Extremely_Loud a dominant saturated structure, so removing the high-scoring region removes the defining evidence. Scattered_Light is an extended horizontal band and Violin_Mode a set of thin distributed lines: random deletion fragments the pattern across the whole image, whereas SAM-guided deletion concentrates removal in one region and leaves the rest — which still carries sufficient evidence — intact. This is a known limitation of deletion-based tests, which penalize explanations of spatially extended features, and it is reported rather than suppressed.

Two controls confirm the measurement itself. All curves converge to approximately 0.25 at full deletion, which is chance for four classes. And the Extremely_Loud random baseline remaining at 0.8892, the highest of the four, is independent confirmation of that class’s behaviour below: its evidence is distributed across the whole image, which is why no localized removal registers.

5.6 Calibrating the divergence metrics

The metrics used to compare two explanations return numbers between zero and one whatever is fed to them. Three questions must be answered before any quantization result can be interpreted: what they report when nothing has changed, when something negligible has changed, and when everything has changed.

All measurements use fifteen validation images per class at layer 2 with $\gamma = 0.5$, on the three checkpoints of Table 5.3.

5.6.1 Verification of the instrument

Comparing a checkpoint against itself returns $\text{PCC} = 1$, $\text{IoU} = 1$ and a centre-of-mass difference of 0 to within 10^{-9} , confirming that the metrics and the image pairing are correct. Two separate calls to the sampling routine return bit-identical image tensors, confirming that every model sees the same data. A second forward pass of the same model reproduces the maps exactly, confirming that the network is deterministic in evaluation mode and that the cross-model differences below are not run-to-run noise.

5.6.2 Two independent solutions

Class	PCC mean	worst	IoU mean	worst	CoM diff, max
Extremely_Loud	0.917	0.869	0.520	0.460	0.076
Violin_Mode	0.844	0.816	0.597	0.543	0.115
Blip	0.612	0.336	0.593	0.503	0.083
Scattered_Light	0.477	0.153	0.376	0.234	0.104

Table 5.6: Divergence between full-precision models differing only in seed, over the three seed pairs. Chance-level IoU at $k = 0.20$ is 0.111.

The spatial metrics collapse. Two models agreeing to within one validation sample in accuracy produce maps correlating at 0.153 on Scattered_Light, with a top-20 % overlap

of 0.234 against a chance level of 0.111. For comparison, a thirtyfold change in γ reduces the correlation only to 0.960: initialization variance is 21 times larger than that effect.

The per-image distribution rules out an outlier explanation. On Blip the fifteen per-image correlations run from 0.147 to 0.768 with a median of 0.282, and twelve of the fifteen fall below 0.35. The disagreement is uniform across samples.

The mechanism is not a defect of the metric. Networks trained from different initializations learn functionally equivalent but **permuted** representations, so a pixel-wise comparison between them compares arbitrary assignments of channels to positions rather than information content. This is why the result is a bound on cross-model spatial claims rather than the threshold for the quantization grid, where the quantized model inherits its parent’s weights and no permutation exists.

The temporal centre of mass survives. The maximum difference across seed pairs is 0.115 time steps, against the ≈ 0.90 produced by a thirtyfold change in γ — a factor of eight. Being an aggregate over the whole map, it is insensitive to which neuron occupies which position, which is precisely what defeats the pixel-wise metrics.

5.6.3 A negligible perturbation

The grid compares a quantized model with its own parent, so the operative threshold is what a perturbation far smaller than quantization already produces. Gaussian noise was added to the weight tensors of one checkpoint — biases and LIF buffers excluded, since perturbing β would confound membrane dynamics with weight quantization — at magnitudes expressed as fractions of the measured 8-bit step of those weights (0.00168, from a maximum absolute weight of 0.214).

This is a graded form of the model randomization test of Adebayo et al. [1]: where they randomize weights to verify that a metric responds at all, this perturbs towards zero to find where it stops responding.

Noise / 8-bit step	Extremely_Loud	Blip	Scattered_Light	Violin_Mode
0.01	1.0000	0.9997	0.9996	0.9991
0.1	0.9999	0.9981	0.9977	0.9930
0.5	0.9998	0.9949	0.9938	0.9824
1.0	0.9997	0.9921	0.9901	0.9749

Table 5.7: Spatial PCC against the unperturbed model. The 4-bit and 2-bit quantization steps are 17 and 85 times larger than the largest perturbation shown.

The spatial metrics are usable after all, for same-parent comparisons. PCC remains above 0.975 under a perturbation as large as the entire 8-bit step. The collapse of Table 5.6 is a consequence of comparing different initializations, not a fragility of the metric; the two results answer different questions.

The metrics differ sharply in dynamic range. Measuring the distance from perfect agreement, PCC starts 0.025 away from 1 and can reach 0.847: 34 times its own floor. IoU starts 0.296 away and reaches 0.766, a factor of only 2.6, and is already at 0.704 under a perturbation 85 times smaller than the 2-bit step. IoU should therefore be read as a sensitive early indicator that a change has occurred, and PCC as the graded measure of how large it is. The temporal centre of mass sits between them, with a factor of 3.9 between its numerical floor (0.029 steps) and the seed level (0.115).

Extremely_Loud does not respond. Its PCC is 0.9997 under the largest perturbation and its worst per-timestep correlation is 0.996. Combined with its behaviour under γ , this class cannot register an explanation change by any measure tested here. A flat grid result for Extremely_Loud will not distinguish absence of effect from inability to detect one, and this is stated in advance.

5.6.4 What the calibration establishes

Level	What it measures	PCC	IoU	CoM
Numerical floor	Weight noise at $1 \times$ the 8-bit step	0.975	0.704	0.029
γ reference	A thirtyfold change in the kernel	0.960	—	0.90
Independent solutions	Two FP32 models, different seeds	0.153	0.234	0.115

Table 5.8: The three calibration levels, worst class in each case, at 112 px, layer 2, $\gamma = 0.5$, on validation.

The *numerical floor* is the operative threshold for the grid: a divergence below what a negligible weight perturbation already produces is measurement noise. The γ *reference* gives an intuition: if quantization produced a PCC of 0.90, it would be doing more than a thirtyfold change of the explainer itself. The *independent-solutions* level is a pessimistic upper bound and a result in its own right.

The class sensitivity ordering — Violin_Mode > Scattered_Light > Blip > Extremely_Loud — recurs identically under three unrelated interventions: changing the kernel, changing the initialization, and perturbing the weights. Explanation sensitivity is therefore a property of the class morphology, and every grid result must be reported per class, since a pooled figure would average a class that responds strongly with one that does not respond at all.

5.7 Preparing the membrane arm

5.7.1 The range of the membrane potential

The membrane potential of every LIF layer was recorded at every time step over the full validation partition (1 435 images, eight steps) on the reference checkpoint. Table 5.9 summarizes the distributions.

Layer	min	p _{0.1}	median	p _{99.9}	max	std	> U_{thr}
1	−12.75	−9.43	−0.08	6.40	8.76	1.98	14.9 %
2	−15.73	−11.10	0.25	5.63	11.64	1.42	15.1 %
3	−47.03	−36.82	0.00	11.27	20.23	4.22	15.3 %
4	−26.04	−17.01	−0.62	6.34	14.67	2.59	11.5 %

Table 5.9: Membrane potential of the four hidden layers over the validation partition, reference checkpoint (seed 1234). The last column is the fraction of values above the firing threshold $U_{\text{thr}} = 1$, which equals the mean firing rate of the layer.

The distributions are wide and strongly skewed towards negative values: in layer 3 the potential reaches −47, almost fifty thresholds below rest. A grid covering the full observed range would be dominated by values far from the threshold, which is the only point at which a potential affects a spike decision. At 4 bits, a symmetric grid over $[-47, +47]$ would have a step of more than six thresholds, and the network could not be represented at all.

5.7.2 Clipping the potential costs nothing

Before any quantization, the potential of the four hidden layers was *clipped* at every time step to a candidate range, on the full-precision network, and the effect on accuracy and

dynamics was measured. Clipping isolates the choice of range from the choice of bit-width: if a range already changes the network in full precision, no grid built on it can be faithful.

Clipping range	Val. accuracy	Δ samples	Layer-4 CoM
none	0.98328	—	3.9469
per layer, $[p_{0.1}, p_{99.9}]$	0.98328	0	3.9469
per layer, $[p_1, p_{99}]$	0.98328	0	3.9469
fixed $[-8, +4]$	0.98328	0	3.9469
fixed $[-4, +4]$	0.98328	0	3.9469
fixed $[-2, +2]$	0.98258	-1	3.9415
unsigned $[0, +4]$	0.98258	-1	3.9483
unsigned $[0, +2]$	0.98258	-1	3.9431
control $[-0.5, +0.5]$	0.84808	-194	4.0684

Table 5.10: Effect of clipping the hidden membrane potentials on the full-precision reference (seed 1234), on validation. The tolerance is the spread of the three-seed reference, four samples out of 1435. The last column is the temporal centre of mass of the layer-4 spike counts.

Three observations follow from Table 5.10.

Negative potentials carry no information the network uses. Clipping to $[0, 2]$ removes every negative value — in layer 3 a range of 47 thresholds — and costs one validation sample out of 1435, a quarter of the spread between seeds. The per-layer dead fractions are unchanged to the first decimal (16.1 / 12.5 / 11.3 / 22.5 % against 16.1 / 12.6 / 11.3 / 22.8 %), and the largest shift in the temporal centre of mass of any layer’s spike counts is 0.004 steps, well below the 0.029-step floor measured for the SAM centre of mass in Section 5.6 (the two quantities are related but not identical). Below rest, how far below no longer matters: a neuron at -20 and a neuron at 0 both need more than one threshold of input to fire.

The upper bound can be tight. Potentials above twice the threshold are rare and can be clipped at no measurable cost. Since the clipping acts after the spike decision, it only alters the carried state of neurons that have just fired well above threshold; the spike they emitted at that step is unaffected.

The clipping is not inert. The control range $[-0.5, +0.5]$, which keeps the potential permanently below threshold after each step, drops accuracy to 0.848 and raises the layer-3 firing rate from 0.153 to 0.227. The hook therefore acts on the state the network actually carries, and the null results above are genuine.

The unsigned range $[0, 2]$ is adopted: it is the tightest range tested that stays within tolerance, it centres the threshold in the grid, and it spends no levels on negative values, which saves one bit at every bit-width compared with the symmetric $[-2, 2]$ that performs identically.

5.8 Energy and memory of the reference

Table 5.11 applies the accounting of Section 4.8 to the full-precision reference, using the per-hidden-layer firing rates averaged over the three seeds (0.258/0.186/0.185/0.126).

The spiking network costs 62.8 μJ per inference, 2.5 times less than a convolutional network of identical shape. The saving depends entirely on the treatment of the first layer: recomputing its constant current at every step would cost 183.9 μJ , more than the convolutional equivalent, and the two readings differ by a factor of 2.9. Even computed once, the first convolution is 28 % of the budget, because it is the only layer that multiplies.

Layer	MAC (dense)	Operations	Energy (μJ)
conv1 (analog input, once)	3.76×10^6	3.76×10^6 MAC	17.31
conv2	10.04×10^6	20.72×10^6 SOP	18.65
conv3	10.04×10^6	14.93×10^6 SOP	13.44
conv4	10.04×10^6	14.83×10^6 SOP	13.34
linear	0.03×10^6	0.03×10^6 SOP	0.02
SNN, conv1 once			62.8
SNN, conv1 at every step			183.9
Equivalent CNN (all MAC)	33.89×10^6		155.9

Table 5.11: Estimated energy per inference of the FP32 reference at 112 px, $T = 8$, with 45 nm costs of 4.6 pJ per MAC and 0.9 pJ per accumulate.

The initialization regime of Section 5.2 reaches the energy as well. Computed seed by seed, the estimate is 50.0, 78.1 and 60.1 μJ for seeds 1234, 2345 and 3456, the difference coming almost entirely from the first two layers, where the firing regime varies most. Three models indistinguishable in accuracy therefore differ by more than half in estimated energy, and every energy comparison in the grid is made at matched seed.

In memory, the 295 088 weights occupy 1 152.7 KiB in 32-bit float (the 244 biases, kept in full precision, are negligible), and the membrane state of the 94 080 hidden neurons another 367.5 KiB. At 8, 4 and 2 bits these become 288, 144 and 72 KiB for the weights and 92, 46 and 23 KiB for the state.

5.9 The quantization grid

All thirty fine-tuning runs completed without failure. Unless stated otherwise, every figure below is a mean over the three seeds, each quantized model being compared with the full-precision parent of its own seed, on validation. Figure 5.9 summarizes the grid; the sections that follow take it apart.

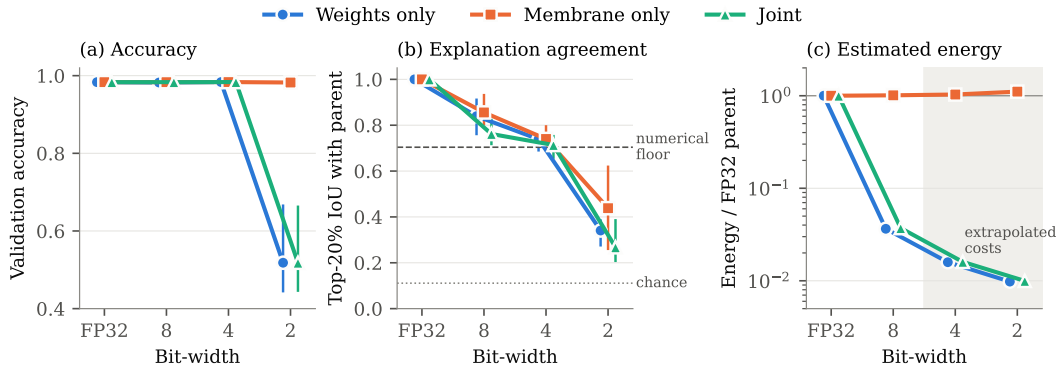


Figure 5.9: The grid on a common bit-width axis. (a) Validation accuracy after fine-tuning. (b) Top-20 % IoU between the SAM maps of each quantized model and those of its parent, mean over the four classes; the dashed line is the numerical floor of the calibration, the dotted line chance. (c) Estimated energy per inference relative to the parent, logarithmic scale; the shaded region uses extrapolated per-operation costs. Points are means over three seeds, bars span the seeds. Markers are offset horizontally for legibility.

5.9.1 Gates and control

The quantizer passed both gates. Attaching and removing it restores the original weights exactly, and a quantized checkpoint reloads with zero drift. With no fine-tuning, the joint 8-bit model of seed 1234 reaches 0.98328 validation accuracy, identical to its parent; the 8-bit weight arm loses three samples out of 1 435, inside the four-sample tolerance.

The control arm, which runs the full fine-tuning procedure with no quantizer, returns a spatial correlation of exactly 1, an IoU of exactly 1 and a centre-of-mass shift of exactly 0 with its parent, in every class and every seed. Fine-tuning at the reduced learning rate never improved on the parent’s validation accuracy, so the selected control checkpoint is the parent itself: the pipeline — reload, fine-tune, select, re-evaluate — introduces nothing when nothing is quantized.

The guard of Section 4.6 holds as well: the 8-bit weight arm keeps a spatial correlation of at least 0.982 with its parent in every class of every seed.

5.9.2 Accuracy

Arm	Bits	PTQ	After fine-tuning	Δ samples	Epochs
Weights	8	0.9819	0.9823 ± 0.0008	−1.3	9.7
Membrane	8	0.9814	0.9833 ± 0.0000	0.0	8.3
Joint	8	0.9819	0.9833 ± 0.0007	0.0	8.0
Weights	4	0.9547	0.9835 ± 0.0016	+0.3	11.7
Membrane	4	0.9807	0.9835 ± 0.0011	+0.3	10.3
Joint	4	0.9508	0.9835 ± 0.0015	+0.3	14.0
Weights	2	0.3591	0.5182 ± 0.1300	−667	18.3
Membrane	2	0.9517	0.9823 ± 0.0004	−1.3	13.0
Joint	2	0.3652	0.5182 ± 0.1275	−667	19.0

Table 5.12: Validation accuracy across the grid: post-training quantization (PTQ, no fine-tuning) and after fine-tuning, mean $\pm$ standard deviation over three seeds. Δ is the difference from the FP32 reference (0.9833 ± 0.0014) in validation samples out of 1 435.

At 8 and 4 bits accuracy is indistinguishable from full precision in every arm (Table 5.12); every difference is below one validation sample on average. Fine-tuning matters at 4 bits: post-training quantization of the weights costs about 40 samples, and fine-tuning recovers all of them.

At 2 bits the arms separate completely. The membrane arm keeps parity, 0.9823, within three samples of its parent in every seed. The weight and joint arms collapse to 0.518, and the collapse is not uniform across classes: no 2-bit weight model ever predicts Scattered_Light, and two of the three never predict Violin_Mode either, assigning every image to Blip or Extremely_Loud. Fine-tuning recovers 16 points over post-training quantization but cannot recover a ternary network. A 2-bit membrane, with a single level between rest and threshold, is enough to classify; ternary weights are not.

The asymmetry inverts the prediction derived from Smith et al. [23] in Section 4.4, that the membrane would need at least four bits while weights would survive two. In this network and with this grid, it is the other way round.

5.9.3 Explanation divergence

At 8 and 4 bits the explanations stay at the floor. Spatial correlation remains above 0.977 on average in every arm, and the IoU between 0.71 and 0.86, around the numerical floor of 0.704: at 4 bits four of the twelve class-and-seed cells of each arm fall marginally below it, the lowest at 0.595, all far above the independent-solutions level of 0.234 (Tables 5.13

Arm	Bits	Blip	Extremely_Loud	Scattered_Light	Violin_Mode	Mean	Worst cell
Weights	8	0.821	0.854	0.841	0.841	0.839	0.716
Membrane	8	0.845	0.876	0.843	0.858	0.856	0.766
Joint	8	0.753	0.779	0.749	0.766	0.762	0.674
Weights	4	0.719	0.787	0.717	0.733	0.739	0.649
Membrane	4	0.726	0.773	0.719	0.743	0.740	0.628
Joint	4	0.696	0.752	0.691	0.715	0.714	0.595
Weights	2	0.269	0.335	0.415	0.344	0.341	0.127
Membrane	2	0.300	0.559	0.490	0.404	0.438	0.102
Joint	2	0.217	0.339	0.344	0.171	0.268	0.057

Table 5.13: Top-20 % IoU between the SAM maps of each quantized model and those of its parent, layer 2, $\gamma = 0.5$, twenty validation images per class, mean over three seeds. “Worst cell” is the lowest class-and-seed value. Numerical floor 0.704, chance 0.111.

Arm	Bits	PCC mean	PCC worst	CoM $ \Delta $ mean	CoM $ \Delta $ worst	CoM signed
Weights	8	0.995	0.982	0.007	0.015	−0.005
Membrane	8	0.995	0.989	0.010	0.025	−0.009
Joint	8	0.989	0.973	0.022	0.049	−0.022
Weights	4	0.982	0.960	0.049	0.094	−0.048
Membrane	4	0.985	0.963	0.044	0.064	+0.043
Joint	4	0.977	0.949	0.021	0.043	+0.018
Weights	2	0.827	0.748	0.211	0.369	−0.211
Membrane	2	0.923	0.799	0.068	0.137	−0.053
Joint	2	0.780	0.580	0.235	0.367	−0.235

Table 5.14: Spatial Pearson correlation and temporal centre-of-mass shift (in time steps) against the parent, same protocol as Table 5.13. A negative signed shift means that the spike energy of the explanation moves earlier. Numerical floors: PCC 0.975, CoM 0.029; independent-solutions level: PCC 0.153, CoM 0.115.

and 5.14). The temporal centre of mass is the only metric that moves at 4 bits, by 0.044 to 0.049 steps in the two single-target arms, above its floor of 0.029 and below the 0.115 that separates independent solutions. The direction is informative: quantizing the weights moves the spike energy of the explanation *earlier*, quantizing the membrane moves it *later*, and in the joint arm the two shifts partly cancel (+0.018).

At 2 bits the weight and joint arms explain a different classifier. Their IoU falls to 0.34 and 0.27 and their centre of mass shifts by more than the independent-solutions level. Since those models no longer classify, the divergence of their explanations is a consequence of the collapse rather than a measurement of explanation fidelity at constant accuracy, and it is not read as such.

The membrane arm at 2 bits is the result the grid was built to find. Accuracy is at parity, yet the IoU with the parent's explanation is 0.438, 38 % below the numerical floor. All twelve class-and-seed cells are below the floor, and two are below chance (the lowest 0.102): for those, the 20 % of the image that SAM highlights in the quantized model overlaps the parent's no more than two random selections would.

The comparison that carries the thesis is between two models with the same accuracy. The 4-bit weight model and the 2-bit membrane model both reach 0.983 (0.9835 and 0.9823). The first explains its decisions with essentially the same pixels as its parent, IoU 0.739; the second shares less than half of them, IoU 0.438. Selected on accuracy, the two are interchangeable; their explanations are not.

The spatial correlation alone would have given the wrong answer. The 2-bit membrane model scores a PCC of 0.923. Against the values that the literature presents as preserved explanations, around 0.85, that figure would be reported as a surviving explanation; it is below the numerical floor, but by an amount that understates what changed. Pearson correlation is dominated by the broad, low-intensity structure that every map of a class shares, while the top-k overlap measures the region a reader is actually shown. This is the disagreement between metrics that Bylinskii et al. [5] warn of, and it is why both are reported.

The effect depends on the seed. The 2-bit membrane IoU is 0.255, 0.624 and 0.435 for the three seeds, a spread larger than any effect measured within a seed below. The class ordering is also not the one of the calibration: Blip, not Violin_Mode, loses the most (0.300), and Extremely_Loud, which registered no perturbation in the calibration, now moves (0.559).

5.9.4 The hypothesis

The hypothesis predicted an ordering: the membrane arm displaces the temporal centre of mass at a higher bit-width than the one at which the weight arm lowers the spatial correlation. The grid does not show that ordering. Both effects first appear at the same bit-width, 4 bits, where the weight arm lowers the PCC below the per-class floor on every class and both single-target arms shift the centre of mass by the same amount. At 2 bits, where the arms do separate, the visible effect of membrane quantization on the explanation is *spatial* — which pixels are highlighted — while its centre-of-mass shift (0.068) stays below the independent-solutions level. **As formulated in terms of SAM metrics, the hypothesis is not supported.**

Its mechanism, however, is visible elsewhere. The per-model validation pass measures the regularity of firing, and there the membrane arm at 2 bits does something no other configuration does.

In every model with a 2-bit membrane, the inter-spike intervals of layer 1 become perfectly regular (Table 5.15): the CV is exactly zero in all six measurements, while it is unchanged with 2-bit weights and with a 4-bit membrane. The fraction of neurons for which the CV is defined is comparable in every configuration, so the effect is not a change in which neurons are measured. The mechanism follows from Table 4.3: with a single level

Configuration	Seed 1234	Seed 2345	Seed 3456	Neurons with defined CV
FP32 parent	0.132	0.241	0.051	2.7 / 24.1 / 10.4 %
Membrane, 4 bit	0.134	0.236	0.051	2.5 / 23.4 / 10.3 %
Weights, 2 bit	0.140	0.104	0.208	4.0 / 25.3 / 10.0 %
Membrane, 2 bit	0.000	0.000	0.000	3.2 / 28.5 / 10.7 %
Joint, 2 bit	0.000	0.000	0.000	5.1 / 27.2 / 10.6 %

Table 5.15: Mean coefficient of variation of the inter-spike intervals in layer 1, per seed, and the fraction of neurons for which it is defined. The values are from the single validation pass of Section 4.6 and are not comparable with Figure 5.3.

between rest and threshold, the state of a neuron between two spikes can take only one intermediate value, so under the constant drive of direct encoding the number of steps from reset to threshold cannot vary, and firing becomes periodic. Membrane quantization does act on the temporal dynamics first, as the hypothesis argued; but the change is in the *regularity* of the spike trains, and the temporal centre of mass, which measures where the spike mass falls in time and not how regularly it arrives, cannot see it.

The other diagnostics behave accordingly. The 2-bit membrane raises the firing rate of every layer (from 0.258/0.186/0.185/0.126 to 0.273/0.218/0.229/0.151) with little change in the dead fraction, whereas 2-bit weights double the dead fraction of layer 4 (from 23.6 to 43.1 %) and saturate up to 6.8 % of layer 1.

5.9.5 Faithfulness

The deletion metric was repeated on the 2-bit membrane models, the only configuration with intact accuracy and a changed explanation, with the baseline recomputed on the parent of each seed (Table 5.16).

Seed	Class	Parent baseline	Own map	Parent's map
1234	Blip	0.153	0.166	0.117
1234	Extremely_Loud	0.255	0.248	0.236
2345	Blip	0.087	0.083	0.099
2345	Extremely_Loud	0.222	0.136	0.149
2345	Scattered_Light	0.056	0.074	0.083
2345	Violin_Mode	0.106	0.187	0.193
3456	Blip	0.111	0.033	0.037
3456	Extremely_Loud	0.207	0.186	0.180
3456	Scattered_Light	0.112	0.094	0.056

Table 5.16: Deletion gap (AUC of random deletion minus AUC of SAM-guided deletion; larger means more faithful) for the 2-bit membrane models, ten validation images per class. Only the nine class-and-seed cells whose parent baseline is at least 0.05 are shown; the other three have no signal to degrade.

The explanation is changed, not broken. In every interpretable cell the quantized model's own map still separates from random deletion; the median gap is 0.95 of the parent's, and only one cell comes close to losing it — Blip on seed 3456, from 0.111 to 0.033. Explaining the quantized model with its parent's maps works as well as with its own: the mean difference between the two gaps is +0.0065 over the nine cells, four positive and five negative, never more than 0.049, although the two maps share only 44 % of their highlighted pixels.

The two measurements are therefore not in contradiction; they answer different questions. The deletion metric rewards any map that points at evidence the model uses, and when the evidence is redundant — a Blip stroke covers many positions, any large part of

which suffices — several disjoint sets of pixels satisfy it. The IoU says which of those sets a reader is shown. A commissioner looking at the quantized model’s map would be sent to a region that does carry evidence, but not to the one the original model would have shown.

Which classes SAM can explain at all depends on the seed. Only nine of the twelve class-and-seed cells have a baseline signal: Violin_Mode is explainable in one parent of three, Scattered_Light in two of three, and even Blip’s baseline gap ranges from 0.087 to 0.153 across networks of equal accuracy. The set of classes on which faithfulness can be assessed is a property of the individual network, not of the class.

5.9.6 Stratification by event strength

Class	Membrane, 2 bit			Membrane, 4 bit		
	1234	2345	3456	1234	2345	3456
Blip	+0.121	+0.038	+0.164	+0.060	+0.012	+0.039
Scattered_Light	+0.443	−0.021	+0.138	+0.242	+0.017	+0.109
Violin_Mode	+0.134	−0.038	−0.124	+0.093	+0.022	+0.028
Mean of the three	+0.233	−0.007	+0.059	+0.132	+0.017	+0.059
Extremely_Loud	−0.469	−0.202	−0.354	−0.234	−0.101	−0.142

Table 5.17: IoU gradient with event strength: IoU in the high-SNR tercile minus IoU in the low-SNR tercile, terciles computed within each class, fifteen images per cell. A positive value means that weak events lose their explanation more.

The data do not support the hypothesis that degradation concentrates on weak glitches. The mean gradient over Blip, Scattered_Light and Violin_Mode is +0.233, −0.007 and +0.059 in the three seeds at 2 bits (Table 5.17): positive on average, but the spread between seeds is as large as the effect, and the between-seed range of the 2-bit IoU itself (0.255 to 0.624) exceeds the largest within-seed gradient.

One effect is consistent, and of opposite sign. On Extremely_Loud the gradient is negative in all six measurements: the *strongest* events lose their explanation most. These are the events that saturate the spectrogram, so that most of the image carries evidence and the top-20 % region is least constrained; a small change of the network is enough to move it.

5.9.7 Energy and memory

The membrane arm does not save energy; it costs more. Its weights stay in full precision, so the cost of each operation never falls, while the quantized membrane raises the firing rates and with them the number of operations: +1.1 % at 8 bits, +3.1 % at 4, +10.8 % at 2 (Table 5.18). What it does save is state memory, 344 KiB at 2 bits, almost a quarter of the total footprint.

Quantization raises the operation count in every arm, mildly at 8 and 4 bits (1–8 %) and sharply at 2 bits, where it grows by half with quantized weights: the collapsed networks fire more, not less. In the weight and joint arms the saving still comes entirely from the cheaper operations.

There is no trade-off between efficiency and explainability in this grid. The configuration that changes the explanation at constant accuracy, the 2-bit membrane, is dominated on every axis by the 4-bit weight model: same accuracy, IoU 0.739 against 0.438, energy 0.016 against 1.108 times the parent, memory 512 against 1 176 KiB. The configuration to deploy is the **joint 4-bit model**: accuracy at parity (0.9835), explanation at the floor (IoU

Arm	Bits	SOP (10^6)	Energy (μ J)	$\times$ FP32	Memory (KiB)
FP32	32	50.50	62.76	1.000	1520.2
Weights	8	51.02	2.28	0.037	655.7
Membrane	8	51.34	63.52	1.011	1244.6
Joint	8	52.29	2.32	0.037	380.0
Weights	4	54.21	1.00*	0.016	511.6
Membrane	4	52.62	64.67	1.031	1198.6
Joint	4	54.84	1.01*	0.016	190.0
Weights	2	74.60	0.61*	0.010	439.5
Membrane	2	57.81	69.34	1.108	1175.7
Joint	2	76.06	0.62*	0.010	95.0

Table 5.18: Synaptic operations, estimated energy and memory per inference, mean over three seeds, from the firing rates of each selected checkpoint. “ $\times$ FP32” is the mean of the per-seed ratios to the parent. *Extrapolated per-operation costs. Memory is weights plus membrane state.

0.714 against 0.704), estimated energy about 1/60 of the parent, and one eighth of its memory (190 KiB).

One assumption decides the size of the saving. The first convolution receives the analog image, and Table 5.18 costs it at the weight bit-width, which assumes the input is quantized to the same width. At 4 bits that layer is 19 % of the joint model’s energy. If it is instead costed in floating point, the joint 4-bit saving factor falls from 62 to 3.5. The operation counts and the memory column do not depend on the assumption, and neither does the conclusion that the membrane arm is dominated; the factor by which the deployable model is cheaper does.

5.9.8 Still to be measured

TO DO. Two analyses need no GPU, the data being already on disk: the Earth Mover’s Distance between the per-neuron firing-rate distributions of each model and its parent (from the `_rates.npz` files; compare with Smith et al. and with the seed-to-seed 0.039), and the disagreement subset (validation images correct in the parent and wrong after quantization, joined on the `index` column of the `_per_sample.csv` files and characterized by their physical descriptors). Add them here or drop the two mentions in Section 4.6 and Section 4.7.

5.10 Test-set evaluation

Model	Test accuracy	95 % Wilson interval
FP32 reference	[–]	[–]
Joint, 4 bit	[–]	[–]
Membrane, 2 bit	[–]	[–]

Table 5.19: Accuracy on the test partition (1 436 images), opened once.

TO DO. Run `evaluate_test.py` once, last, on the three models above (the reference, the recommended deployment configuration, and the configuration with the changed explanation). Add per-class recall.

Chapter 6

Discussion and Conclusions

6.1 The instrument

This thesis set out to measure whether a quantized spiking neural network reasons like its full-precision parent, and found that the measurement could not be interpreted until the instrument was calibrated.

A four-block convolutional spiking network classifies four families of transient noise in LIGO data at 0.9833 ± 0.0014 validation accuracy over three seeds, under constraints — shift-implementable membrane decay, at most eight time steps, 112-pixel input — that keep the configuration deployable by construction rather than by subsequent rounding. Each of those constraints was shown to cost nothing measurable in accuracy, and the resolution choice improved the health of the deepest layer by 12.7 percentage points of dead fraction.

The Spike Activation Map was implemented, its summation range corrected, and its free parameter fixed on the temporal axis it governs rather than on time-aggregated maps that cannot see it. Its faithfulness was tested quantitatively and found to hold on two classes of four, with a geometric explanation for the other two.

The calibration established three levels: the divergence produced by weight noise as large as an entire 8-bit quantization step (PCC 0.975), by a thirtyfold change in the explainer (PCC 0.960), and by initialization alone (PCC 0.153). The third is far larger than expected, and it carries three consequences.

Spatial metrics cannot compare independently trained models. Two networks that agree to within one validation sample in accuracy produce explanations correlating at 0.153. The mechanism is representational permutation rather than any defect of the metric, and the disagreement is uniform across samples rather than driven by outliers.

They can compare a model with its own parent. Under a perturbation 85 times smaller than the 2-bit quantization step, PCC remains above 0.975. The two results answer different questions, and the distinction must be made explicitly whenever such a number is reported. In this thesis it decided the design of the grid: every quantized model is fine-tuned from its own parent.

Published figures are not interpretable without these floors. Values around 0.85 are presented in the literature as evidence of preserved explanations. For this architecture that value falls between the numerical floor and the independent-solutions level, so it cannot be read without knowing which comparison produced it. Establishing the floors requires no training runs beyond a multi-seed baseline that is good practice regardless.

A fourth observation cuts across all of them: the class sensitivity ordering is identical under three unrelated interventions. Explanation sensitivity is a property of the class mor-

phology, not of the perturbation applied, and Extremely_Loud — saturated and spatially extensive — registers no change under any measure tested.

The preparation of the membrane arm added a result of its own: the network makes no use of negative membrane potentials. Clipping the hidden state to $[0, 2]$, which removes up to 47 thresholds of negative range, costs one validation sample and leaves the dynamics unchanged, so the membrane can be stored on an unsigned grid at one bit less than a symmetric one would need.

6.2 Quantization and explanation fidelity

Accuracy is not a sufficient criterion for spiking networks either. The finding of Rabbi et al. [19] on convolutional networks carries over, in a sharper form. Two configurations of this network classify equally well, a 4-bit weight model and a 2-bit membrane model, and their explanations differ by a factor of almost two in agreement with the original. Anyone selecting a configuration on accuracy alone could have chosen the second and deployed a model that reaches the right labels while showing a different part of the spectrogram. In the detector-characterization setting of Chapter 1, that is the scenario that matters: the label is correct, the diagnostic cue is not the one the original model would have given.

The divergence is real but bounded. The changed explanations are still faithful: deleting the pixels they highlight degrades the model's confidence almost as much as deleting the parent's highlights does, and the quantized model can be explained with its parent's maps as well as with its own. The quantized network has not stopped using spectrogram evidence; among several redundant sets of evidence it now highlights a different one. A faithfulness test alone would have declared the explanation intact; an overlap test alone would have suggested it was lost. Both are needed to say what happened.

The hypothesis was wrong in its form and right in its mechanism. The prediction that the membrane would move the temporal axis of the explanation before the weights moved its spatial axis is not supported: both arms first register at 4 bits, and at 2 bits the membrane changes where the explanation points more than when. What the membrane does change first is the temporal dynamics of the network itself: at 2 bits, with a single representable level between rest and threshold, the first layer fires with perfectly regular intervals. The temporal centre of mass, built to capture the time axis of the explanation, is blind to this, because it measures where the spike mass falls, not how it is spaced. A temporal explanation metric for spiking networks needs to see regularity as well as timing, and the present instrument does not.

The comparison with Smith et al. inverts. Smith et al. [23] report that the membrane needs at least four bits while learned weights survive two. Here the membrane survives two bits in accuracy and the weights do not. The difference has a plausible source in the design: the membrane grid of this thesis was fitted to a measured range and made unsigned after a clipping study, which concentrates all of its levels where the network uses them, whereas a ternary weight quantizer has no comparable freedom. The result does not contradict their mechanism — at 2 bits the membrane does change the network, in its regularity and in its explanations — but it shows that the bit-width at which membrane quantization breaks a network depends on how the grid is fitted.

Weak events are not where the explanation fails. The expectation that quantization would disturb spike decisions mostly in weak events, whose potentials sit near threshold, finds no consistent support: the gradient with event strength changes sign across seeds. The only consistent effect is reversed: the most saturated Extremely_Loud events lose

their explanation most, because their evidence is spread over the whole image and the highlighted region is least constrained.

Efficiency and explainability do not trade off here. The configuration that changes the explanation at constant accuracy is the 2-bit membrane, and it is the one that saves no energy: with weights in full precision the per-operation cost stays the same while firing rates rise. The deployable configuration, joint 4-bit, keeps accuracy at parity and the explanation at the numerical floor, at about one sixtieth of the estimated energy and one eighth of the memory. The divergence between the three curves of Figure 5.9 exists, but it occurs at a point no one would choose for efficiency.

Seed variance is the dominant source of variation. Three full-precision networks of equal accuracy differ by a factor of 2.6 in the firing rate of their first layer, by more than half in estimated energy, and in which classes SAM can explain faithfully at all. The 2-bit membrane IoU ranges from 0.26 to 0.62 across seeds. Every comparison in this thesis was made within a seed, and a single-seed study of the same question could have reached any conclusion from “the explanation is preserved” to “it is lost”.

6.3 Limitations

Generalization of the thresholds. The three calibration levels are specific to this architecture, this dataset and this explainer. The *mechanism* behind the collapse, representational permutation between independently trained networks, is general and well documented; the *magnitude* is not, and a different architecture could place the floors elsewhere. What generalizes is the claim that the floors must be measured, not the values measured here.

Labels are model output. Accuracy is agreement with the Gravity Spy convolutional classifier rather than with physical ground truth. A model that reproduced that classifier’s systematic errors would score well here. The per-class confidences are available and would permit stratifying accuracy by the reference classifier’s own certainty, which was not done.

The partition is random with respect to time. If glitches of the same class are correlated within observing periods, a random test set overestimates generalization to new periods. A block split by observing run would be the more conservative validation; the timestamps were dropped during feature selection for the reasons given in Chapter 3 and would have to be recovered.

T sits at the imposed ceiling. The search was forbidden from exploring more than eight time steps, so it is not known whether it would have preferred a longer integration window. An unconstrained search selected seven, which bounds the likely cost but does not eliminate the question.

The deletion metric fails on two classes. The geometric explanation offered in Section 5.5.2 — that deletion-based tests penalize spatially extended evidence — is an interpretation, not a demonstration. An insertion curve, which rewards rather than penalizes distributed evidence, would test it directly.

The explanation is class-agnostic. SAM does not use labels, so the deletion metric measures faithfulness to the model’s *predicted* class. This is the correct reading of the metric and matches its original formulation, but it means misclassified examples are informative and should be inspected separately rather than pooled.

The temporal metric is partly kernel arithmetic. The γ analysis showed that the centre of mass of the SAM energy is dominated by a minority of near-continuously firing neurons. Between models compared at the same γ this component cancels, but it means that the CoM is most sensitive to changes in that minority and may under-report changes elsewhere.

Quantization is simulated. Fake quantization reproduces the values an integer implementation would hold, not its arithmetic: the membrane is quantized as a stored state after each spike decision, while on integer hardware the comparison with threshold would itself be performed on integers. The energy figures are analytical, at a 45 nm reference node, and below eight bits they rest on an extrapolation of the per-operation costs.

Sample sizes of the explanation analyses. Divergence is measured on twenty validation images per class, faithfulness on ten, stratification on fifteen per class and tercile, and the membrane range on a single checkpoint. These are sufficient for the effects reported here, which are large, but limit the resolution with which small differences between configurations can be ranked; the faithfulness analysis covers only the 2-bit membrane arm.

One grid, one fitting procedure. Each bit-width was fitted once per seed, with one fine-tuning schedule and one membrane range. The collapse of the 2-bit weight arm and the survival of the 2-bit membrane are properties of these quantizers; a different weight quantizer (learned scales, a wider grid for the readout) could move the weight arm's breaking point.

The control does not isolate the fine-tuning. Fine-tuning without a quantizer never improved the parent's validation accuracy, so the selected control checkpoint is the parent itself. The control therefore shows that the pipeline is inert, but not what the extra epochs alone would do to a model that does change; the quantized checkpoints combine both effects.

Checkpoints are selected on the validation partition on which they are then analyzed. With selection by accuracy and analysis of explanations, the dependence is weak, but the test-set evaluation is the only fully independent estimate.

The saving factor rests on an assumption. Costing the analog first convolution at the weight bit-width gives a joint 4-bit saving of 62; costing it in floating point gives 3.5. The operation counts, the memory and the ranking of configurations do not depend on it.

6.4 Future work

Deployment. Hardware deployment was deliberately left out of scope. The natural continuation is to export the selected integer configuration to a neuromorphic platform or a microcontroller, and to measure on it the latency, energy and memory that are estimated analytically here.

Normalization. The baseline contains no batch normalization, because quantizing its parameters is a problem distinct from quantizing weights and potentials. Deeper spiking networks need it, and both approximating it with shifts and removing it altogether through progressive training are available; repeating the study on a normalized architecture would test whether the conclusions survive.

The temporal axis of the input. Under direct encoding every temporal structure is generated by the network. Feeding the spectrogram column by column, so that time in the input corresponds to time in the signal, would give the temporal explanation a physical meaning and make the temporal metrics measure something a detector scientist can read.

Wider benchmark. Extending the four classes to the full Gravity Spy taxonomy, with a split by observing run, would make the classification results comparable with the published benchmark.

6.5 Closing remark

Compressed models are increasingly deployed where their explanations are acted upon. This thesis asked whether a quantized spiking network still decides for the same reasons as the network it came from, and found that the question has two prerequisites: knowing what the comparison metrics report when nothing has changed, and comparing every model with its own parent. With both in place, the answer for this network is that it mostly does — down to four bits, in every arm — and that the one configuration that does not is invisible to accuracy, partly invisible to correlation and to faithfulness, and visible to a metric that asks which pixels a reader is shown. That is the configuration a deployment chosen on accuracy could have picked, and the reason the question was worth asking.

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Appendix A

Additional figures

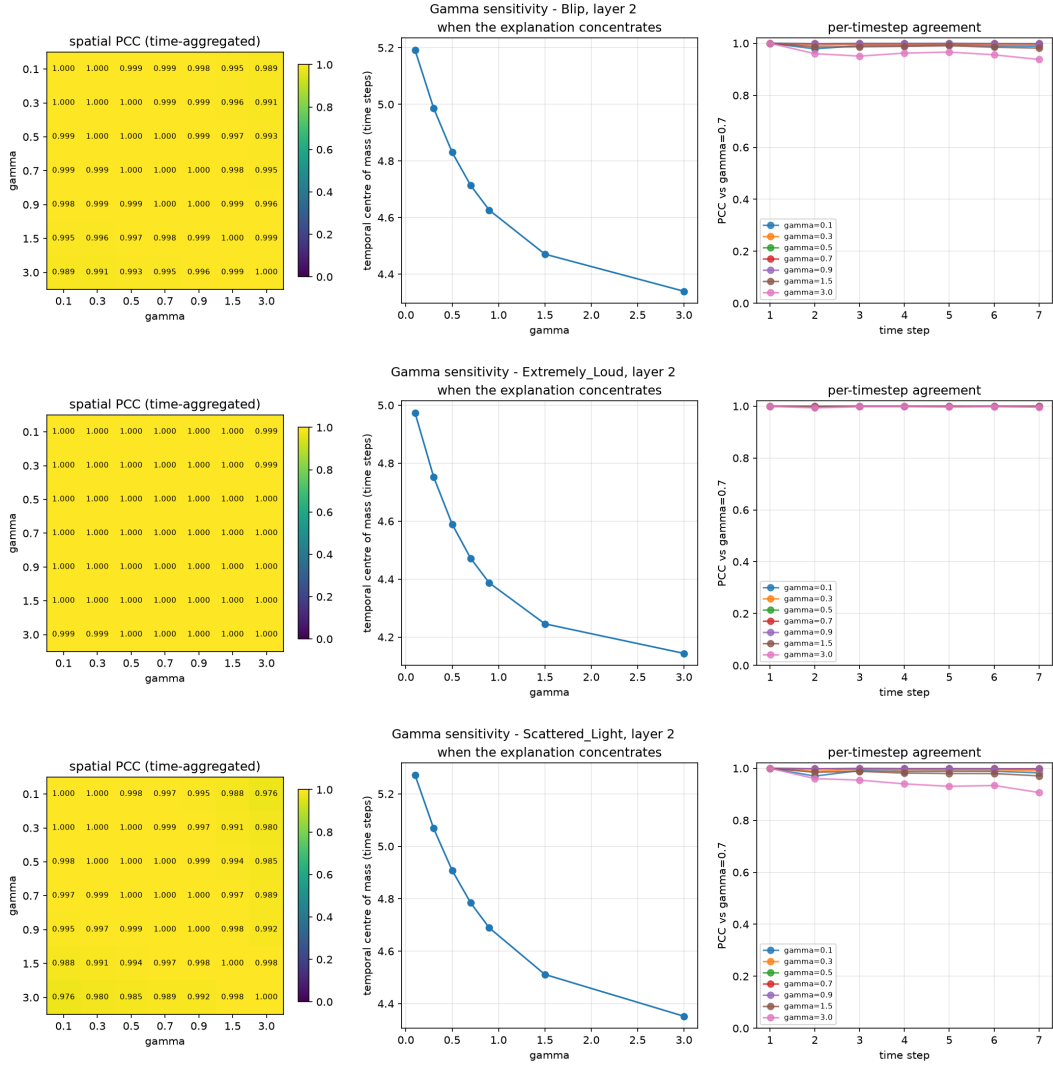


Figure A.1: γ sensitivity at layer 2 for Blip, Extremely_Loud and Scattered_Light (Violin_Mode is in Figure 5.7).

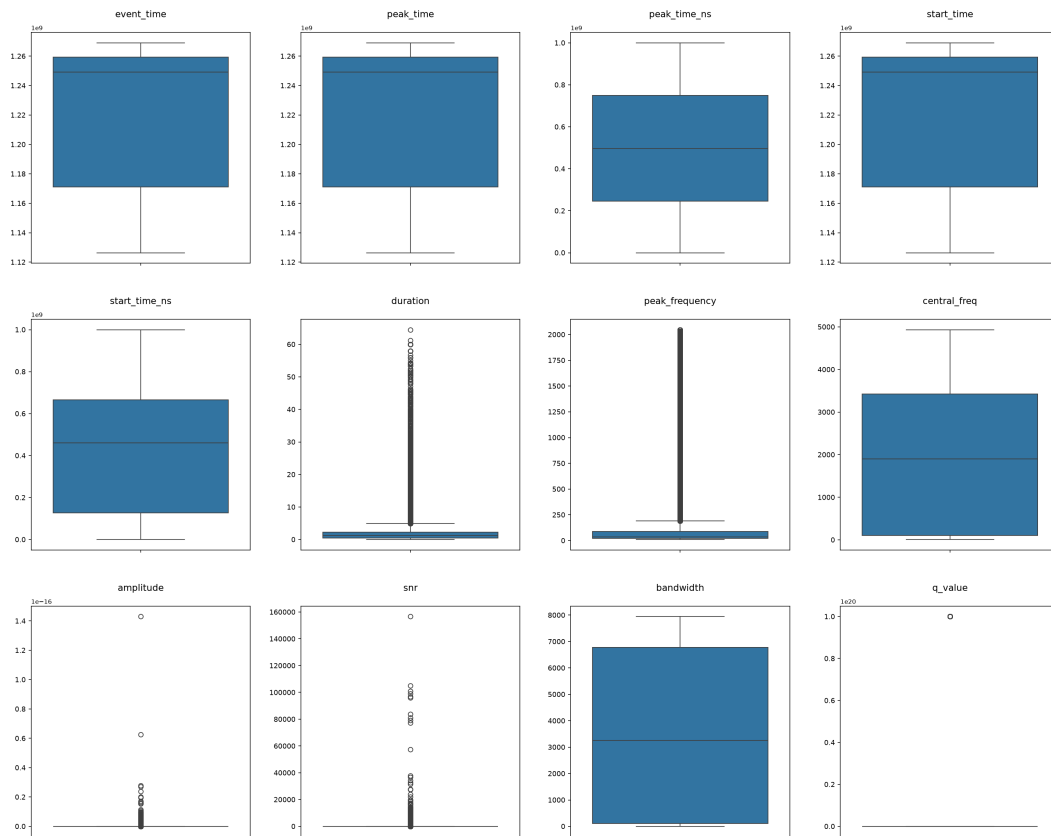


Figure A.2: Distributions of the twelve numerical Omicron descriptors over the full release. The quality factor collapses to a sentinel value of order 10^{20} ; amplitude and SNR are dominated by a long tail of outliers.

Appendix B

Reproducibility

All code, configuration and result files are in the public repository

<https://github.com/devs-glitch/Tesi>

Table B.1 maps each result of this thesis to the script that produces it and the file it writes. Every script reads the model and the partition from `config/baseline_manifest.json`.

Result	Script	Output
Class selection, features	<code>scripts/data_prep.py</code>	<code>selected_dataset.csv</code>
Frozen partition	<code>scripts/freeze_split.py</code>	<code>split_assignment.csv</code>
Hyperparameter search	<code>scripts/tune.py</code>	<code>config/best_hyperparams.json</code>
Resolution study	<code>scripts/resolution_study.py</code>	<code>results/resolution_study.csv</code>
FP32 reference	<code>scripts/fp32_reference.py</code>	<code>config/baseline_manifest.json</code>
Dead neurons	<code>scripts/dead_neurons_check.py</code>	<code>results/dead_neurons.json</code>
Physical validation	<code>scripts/physical_validation.py</code>	<code>figures</code>
γ sensitivity	<code>xai/gamma_sensitivity_analysis.py</code>	<code>results/gamma_sensitivity.json</code>
Deletion metric	<code>xai/deletion_metric.py</code>	<code>figure</code>
Independent solutions	<code>xai/noise_floor.py</code>	<code>results/noise_floor.json</code>
Numerical floor	<code>xai/perturbation_floor.py</code>	<code>results/perturbation_floor.json</code>
Membrane range	<code>scripts/membrane_range.py</code>	<code>config/membrane.json</code>
Clipping study	<code>scripts/test_clipping.py</code>	<code>results/clipping_test.json</code>
Quantizers	<code>model/quantization.py</code>	—
PTQ gate	<code>scripts/ptq_check.py</code>	<code>results/ptq_check.json</code>
Fine-tuning grid	<code>scripts/grid_runner.py, train_quantised.py</code>	<code>results/grid_summary.json</code>
Per-model validation pass	<code>scripts/evaluate_model.py</code>	<code>results/grid/*.json, *.csv, *.npz</code>
SAM divergence	<code>xai/divergence_metrics.py</code>	<code>results/divergence.json</code>
Cross-faithfulness	<code>xai/cross_faithfulness.py</code>	<code>results/cross_faithfulness.json</code>
SNR stratification	<code>xai/stratify.py</code>	<code>results/stratified.json</code>
Energy and memory	<code>xai/energy.py -rates evaluated</code>	<code>results/energy.json</code>

Table B.1: From result to script.

TO DO. Add the scripts for the Earth Mover’s Distance, the disagreement subset and the concluding figure once they are in the repository, and check every path against the final layout.